Dreams as Predictive Neural Rendering

Memory Recombination, Internal Simulation, and State-Dependent Testing

Micah Blumberg
Self Aware Networks Research Institute
selfawarenetworks.com

Final-preprint candidate - bounded synthetic theory-and-methods release - August 13, 2026

Abstract

Dreams present a tractable problem for consciousness science: during sleep, a person may inhabit a vivid, affective, spatially organized scene, recognize objects, converse, choose actions, and experience a viewpoint even though ordinary external sensory constraint and motor execution are altered. A scientific account must therefore explain more than the occurrence of imagery. It must explain how memory fragments are selected and recombined, how a viewpoint and body model are maintained or transformed, how scene transitions are constrained, why internally generated events are ordinarily attributed to an external dream world, how lucidity changes that attribution, and how any of these processes can be measured without reducing the dream to a delayed verbal report.

This paper develops a source-faithful Self Aware Networks (SAN) model called **Predictive Neural Rendering**. The vocabulary is earned from the problem: distributed neural populations contribute partial sensory, mnemonic, affective, spatial, and action-related relations; the current sleep state changes which relations are available and how strongly external evidence constrains them; recurrent exchange builds a situated scene; receiver-specific departures from expected activity update that scene; and the resulting internal consequences constrain what can occur next. Neural Array Projection Oscillation Tomography (NAPOT) names the proposed receive-transform-project reconstruction process. A Phase Wave Differential (PWD) names a candidate receiver-relative departure with timing, phase, waveform, spatial, and consequence dimensions - not every phase difference and not a synonym for prediction error.

The model is placed beside established findings: dream reports occur in REM and NREM sleep; posterior cortical activity predicts reported experience; visual categories can be decoded during sleep onset; dreamed movement recruits sensorimotor cortex; lucid dreamers can signal and answer questions during verified REM; sleep cues can selectively reactivate memories; dream incorporation sometimes predicts later performance; and bodily self-location can be altered experimentally. These findings neither require one universal dream rhythm nor make recalled dream content necessary for memory consolidation. The paper supplies a state-conditioned formal model, identifies competing explanations, and proposes preregistered experiments combining controlled awakenings, closed-loop cueing, lucid signaling, source-attribution reports, high-density electrophysiology, content decoding, and causal perturbation. The purpose is not to declare dreams solved, but to turn a long SAN genealogy - from a 2011 predictive multisensory picture, through 2017-2018 feedback and recurrent-rendering accounts, 2021 memory-replay and lucid-interface notes, and the 2022 NAPOT/PWD architecture - into a testable research program.

The executable program culminates in a preregistered probabilistic latent-mixture gate. On fresh synthetic seeds, the aligned application reaches median exact accuracy 0.8615, median modal-ceiling attainment 1.0017, complete top-two support coverage, and median negative log likelihood only 0.0021 nats above the registered generator

entropy. A matched PWD ablation yields a 0.6420 median exact difference, while independent PWD-null differences remain zero. These are generator-aligned synthetic application results, not human dream data or biological confirmation.

Keywords: dreaming; predictive processing; neural rendering; memory reactivation; lucid dreaming; source attribution; self-location; NAPOT; phase-wave differential; Self Aware Networks

Application development synopsis

Draft 3 adds the mathematical condition that the earlier proposal requires: a latent scene variable is not scientifically interpretable merely because a model can reconstruct observations. Source attribution and vividness become separately recoverable only under declared anchors and a full-rank conditional observation map. PWD parameters become recoverable only from the part of the candidate feature family that remains after projection onto the strongest capacity-matched phase, power, waveform, and prediction-error alternatives. Deterministic synthetic fixtures demonstrate recovery under those conditions and require refusal under collapsed anchors, conditional collinearity, zero PWD effect, participant leakage, and not-missing-at-random reports. These are code and model checks, not human dream evidence.

Draft 4 turns the proposed computational experiment into an executable reference application with a typed transition-event schema, two sequentially learned worlds, twelve predeclared replay and feature arms, eight simultaneously reported outcomes, deterministic hash-chained replay, ten active refusal tests, a leakage audit, a true PWD-null control, and a closed feature-only future final-test partition. In the positive synthetic fixture, receiver-conditioned predictive rendering reaches 0.775 exact next-state accuracy and 0.988 World A retention, but ties exact replay. Its hybrid accuracy is only 0.338, its impossible-transition rate is 0.225, and source calibration ECE is 0.115. Those adverse results remain visible and keep the later performance gate open. The application validates an experimental and software contract, not human dreaming, biological PWD, causal neural perturbation, or artificial consciousness.

Draft 5 freezes a development-only repair before outcomes and tests it across ten new seeds. Training-only constrained decoding removes impossible primary-arm transitions; training-only calibration reduces median source ECE to 0.00078; and median hybrid accuracy rises to 0.756. Yet the strongest factorized receiver-aware comparator is perfect on every development event while using about 3.9 KB rather than roughly 2.5 MB, and the nominal PWD ablation produces a spurious median gain of 0.299 under a PWD-invariant generator because its constraint learner changes too. Row splitting remains severely optimistic. These findings keep the future partition closed and require a harder generator plus a true one-variable PWD ablation before any final-test opening.

Draft 6 supplies that harder development family and a true one-variable PWD comparison. Four independently frozen synthetic rule families add cross-component interactions, receiver-dependent sign reversal, exposed sequence history, distribution shift, and declared hidden ambiguity. The old sixty-key factorized receiver architecture no longer saturates the task, reaching median exact accuracy 0.040. A relational primary arm reaches 0.128 and shows a median full-versus-PWD-ablated difference of 0.038, but it fails the frozen absolute accuracy, hybrid, and transition-validity gates. The matched PWD-null comparison also misses its strict maximum bound on at least one seed. These are development diagnostics, not evidence for biological PWD or dream physiology, and the existing future partition remains unopened.

Draft 7 tests whether a registered PWD-null calibration can remove that specificity failure while a richer estimator improves the hard families. Six new PWD-null fixtures select the largest null-safe multiplier before eight disjoint positive and eight disjoint null fixtures are scored. The selected full multiplier of 1.0 passes both registered and independent null bounds, but the estimator reaches only 0.036 median exact accuracy, performs below the retained Draft 6 relational model and the compact factorized baseline, and produces a median positive PWD difference of only 0.0028. The null-calibration method works for this estimator; the estimator itself does not recover the planted mechanism. Draft 7 therefore remains a failed development architecture, preserves every adverse result, and leaves the future partition closed.

1. The ordinary experience that creates the scientific problem

You can wake from a dream with the memory of having stood in a room that never existed. It had depth. Objects occupied places. Someone may have spoken. You may have moved, felt urgency, made a choice, or believed an impossible transition without noticing its impossibility. The scene was not merely a list of propositions. It was experienced from somewhere.

This ordinary fact creates a demanding engineering question. If the sleeping brain receives less reliable evidence from the immediate environment and blocks most skeletal-muscle output, how does it still construct a world with apparent continuity, perspective, objects, affect, and possible action? A useful theory must explain at least seven operations:

1. selection of memory and semantic material;
2. recombination rather than literal playback;
3. construction of spatial and bodily perspective;
4. prediction of locally plausible scene transitions;
5. attribution of internally generated content to the experienced world;
6. state-dependent access to metacognition and motor channels; and
7. storage, distortion, or loss of the experience when the person awakens.

These operations are related but not identical. Sleep may reactivate a memory without producing a reportable dream. A person may dream without later recall. A lucid dreamer may recognize the state while the scene continues. A report may be reconstructed after awakening. Treating all of these as one event would erase the problem that a mechanism needs to solve.

2. What established dream science already provides

2.1 Dreaming is not identical to REM sleep

Rapid-eye-movement sleep supplied the first strong physiological route into dream research [1,2]. REM sleep remains important: it includes characteristic eye movements, muscle atonia, and a distinctive brain state. Human PET work found increased REM-associated activity in pontine, thalamic, limbic, and selected cortical regions, accompanied by decreases in dorsolateral prefrontal and other regions [3]. Yet reports of experience also occur in non-REM sleep. High-density EEG comparisons found that reported dreaming in both REM and NREM was associated with local reductions in low-frequency activity in posterior cortex, while higher-frequency activity covaried with aspects of reported content [4].

The useful conclusion is not that one stage or one frequency "is" dreaming. Dream availability depends on a distributed, state-dependent configuration. Sleep stage changes the probability, physiology, accessibility, and reported character of experience, but the presence of a particular stage is not a complete content mechanism.

2.2 Internally generated content can engage content-relevant systems

Dream contents are private, but they are not experimentally unreachable. Machine-learning models trained on fMRI patterns predicted reported visual categories during sleep onset above chance [5]. Related work found hierarchical feature overlap across seen and imagined objects [6]. In lucid REM sleep, predefined dreamed hand movements recruited corresponding sensorimotor regions [7], and smooth pursuit of dreamed targets resembled waking perception more than waking imagination in a bounded visuomotor test [9].

These results support a precise proposition: internally generated experience can recruit some of the representational and control machinery used during perception and action without being driven by the same immediate retinal or muscular conditions. They do not show that a complete dream movie has been decoded. They provide partial readouts of a distributed construction.

2.3 Lucidity exposes source attribution and metacognitive access

Ordinarily, dream events are accepted as occurring in the experienced world. Lucidity changes that relation: a dreamer recognizes that the current scene is a dream while remaining asleep. Early EEG/fMRI case evidence implicated additional prefrontal, parietal, and occipitotemporal recruitment during lucid compared with nonlucid REM sleep [8]. Larger and more recent work refines the electrophysiological picture rather than reducing it to a single band.

Most importantly for experimental design, lucid dreamers can use prearranged eye movements and other outputs to time-stamp dream events. Four laboratories demonstrated two-way communication during verified REM sleep: some dreamers received questions, computed answers, and returned distinguishable signals [10]. More recent work used sniff patterns to report dream-eye position and visual-content state in real time. The study recorded 150 signals across 19 sessions from 11 individuals; dream-eye closure did not reliably reproduce the waking alpha increase, and visual-content comparisons were possible only in a three-participant subset [11]. The result is therefore a method demonstration and a bounded physiological observation, not a population-level dream decoder.

A pooled high-density EEG analysis across laboratories also found that lucid REM was not characterized by one simple sensor-level spectral signature. Source-level contrasts included reduced beta power in right central and parietal areas, increased alpha connectivity relative to nonlucid REM, and increased gamma-range power or connectivity around initial lucidity signaling [32]. These regional, temporal, and contrast-dependent results motivate multivariate state and event models; they do not license a universal lucid-dream frequency.

Lucidity therefore turns source attribution into a manipulable variable. The internally generated scene can persist even after the system correctly classifies its source. That dissociation is central to the model developed below.

2.4 Sleep reactivates memory, but dreaming and consolidation are not synonyms

Targeted memory reactivation provides causal evidence that sleep can selectively revisit learned material. Re-presenting an odor context during slow-wave sleep improved hippocampus-dependent declarative retention and recruited the hippocampus [12]. Auditory cues selectively strengthened associated spatial memories [13]. Dreaming about a recently learned navigation task was associated with larger performance improvement in an initial study [14] and an overnight replication [15]. Material-specific memory reprocessing has also been decoded during sleep [16], while nested slow oscillations, spindles, and ripples provide a timing architecture for human sleep-memory exchange [17,18].

The correct inference is bounded. Memory reactivation and consolidation can occur without a recalled dream. Dream incorporation may be a reportable manifestation of a process that also causes later improvement, not necessarily the causal agent of that improvement. SAN needs a model in which memory reactivation can contribute to dream construction without requiring all reactivation to become conscious content.

2.5 Affect, creativity, and self-location are testable dimensions

Dream fear recruited insular and midcingulate systems also involved in waking affect, and individual dream-fear measures covaried with later waking responses [19]. REM sleep improved remote-associate performance in a controlled study [20]. Targeted incubation of an N1 theme was associated with subsequent creative performance, although theme specificity and expectancy remain important questions [21]. These results justify experiments on affective rehearsal and recombination; they do not establish one universal function for every dream.

The apparent location of the self is also experimentally alterable. Focal stimulation near the right angular gyrus induced changes in body perception and viewpoint in a clinical case [22]. Controlled visual-tactile conflict shifted self-identification and self-location in healthy participants [23,24]. Dreamed rooms, bodies, and out-of-body perspectives can therefore be decomposed into measurable variables - viewpoint, self-location, body ownership, sensory congruence, and source attribution - without assuming either that every report has the same cause or that the experience was unreal to the person having it.

2.6 Phenomenology requires structured measurement, not one vividness score

A 2026 validation study developed a REM-to-waking comparison scale with 35 items across eight dimensions: metacognitive awareness, cognitive fluency, memory, temporal awareness, spatial awareness, emotional awareness, perceptual and bodily senses, and interoceptive and chemical senses [33]. The scale was derived and tested in online samples rather than simultaneous sleep-laboratory recording. It is therefore useful as an item and construct source, not as proof that retrospective questionnaires recover the complete real-time dream state. The experiments below combine such dimension-specific reports with immediate awakenings, physiological event anchors, signal confidence, and report-delay measures.

3. The SAN genealogy of the problem

The current formulation did not appear all at once. It developed through a sequence of public and date-bounded stages, each adding a different operation.

3.1 2011: the predictive multisensory picture

An internally dated May 2011 SAN ancestor described distributed "pixels" contributing different parts of a changing scene, learned statistical relations supporting prediction, and the senses joining a continually updated multisensory picture. The same note used the mind-generated-dream metaphor. Its later public Git fixation is preserved separately from the internal date. This source establishes the predictive-rendering ancestry, not the later sleep-dream mechanism or NAPOT vocabulary.

3.2 2017: feedback, hallucination, and recurrent traffic

In the June 2017 Neural Lace Podcast conversation with Android Jones, Micah Blumberg connected Google DeepDream-style feedback amplification, altered imagery, nested neural feedback loops, thalamocortical traffic, meaning construction, and the possibility that excessive recurrent processing could fill gaps or stabilize internally generated forms. The associated article also described EEG-driven virtual environments as mirrors through which people could learn relations between thoughts, emotions, and brain activity.

This stage contributed two lasting ideas. First, imagery depends on what a recurrent system amplifies and interprets, not only on what enters through the senses. Second, feedback can become part of the experienced environment and change the system that generated it.

3.3 2018: render, process, and render again

In the September 2018 Neural Lace Podcast conversation with Jules Urbach, Blumberg described the brain as constructing a light-field-like scene and proposed that a mind repeatedly renders, processes what was rendered, and renders again. That is the clearest public pre-NAPOT ancestor of recurrent neural rendering. It is not a claim that literal pixels or a graphics card exist inside the head. It proposes a function: distributed neural transformations repeatedly produce content that becomes input to later transformations.

3.4 2021: lucid interfaces, memory replay, and the personal holodeck

The 2021 Google Recorder corpus separates three parts of the later dream theory. One recording proposed a computer-facilitated lucid-dream world and described an EEG-to-light-and-sound loop in which the brain receives transformed evidence of its own activity. Another proposed that one memory "pings" related memories and that replay follows learned excitatory and inhibitory routes, making recurrence constrained rather than purely random. A third described the brain as a "personal holodeck" that constructs an organism-relative world and used point-cloud and semantic-segmentation analogies to ask how objects and spatial relations become internally available.

A separate recurrent-flow dream described consciousness as a changing loop or attractor. It remains a first-person hypothesis generator. Its scientific value lies in the questions it prompted, not in treating a dream report as proof.

3.5 2022: NAPOT and PWD

The 2022 SAN repository named the receive-transform-project architecture **Neural Array Projection Oscillation Tomography**. Different populations contribute partial relations; a receiving population transforms incoming activity according to its learned morphology and current state; recurrent exchange constrains later patterns; and no localized inner viewer is required.

The repository later named a **Phase Wave Differential**: a receiver-relative departure from an expected or tonic relation that can include changes in timing, phase, frequency, duration, amplitude, spatial recruitment, excitation, or inhibition. The PWD matters only if a receiver is tuned to it and it changes later processing. This distinction prevents the dream model from equating content with undifferentiated "brain waves."

3.6 2024-2026: the explicit dream research program

SAOv9 explicitly organized dreams around predictive simulation, memory consolidation, formation, choice, sleep cycles, affect, creativity, problem solving, and 3D memory-prediction rendering. The 2026 SAOv10 atomization adds the controls that make those topics scientific: REM/NREM/lucid/waking stratification; cueing and awakening protocols; viewpoint and self-location; source attribution; report delay; base rates; waking-incubation controls; blinded scoring; and falsification.

The present paper gives those components one operational architecture while preserving their dates. Its equations and experiment designs are 2026 formalizations, not retroactively inserted into earlier sources.

4. Earning the term Predictive Neural Rendering

Before naming the mechanism, consider what the sleeping brain must do.

It begins with a learned system whose synapses, dendrites, recurrent connections, and body-state pathways constrain what can be reactivated. Its current sleep state changes neuromodulation, external-input precision, effective connectivity, and motor access. Partial memories and semantic relations become active. Each receiving population transforms those inputs according to its own state. Recurrent exchange stabilizes some combinations, suppresses others, and produces a situated scene. That scene supplies the context in which the next event is interpreted. The process repeats.

This is predictive in a local and testable sense. If a dream includes a hallway, the current model makes some continuations more likely than others: forward movement, a doorway, a turn, another room, a person expected at the destination. The prediction can be bizarre relative to waking physics while remaining locally constrained by the dream's active memory and semantic structure. The term does not claim that the dream reliably forecasts an external future event.

This is rendering in a functional sense. Distributed activity produces a scene that the same distributed system uses to guide attention, expectation, emotion, and imagined action. No homunculus views an internal display. The system "reads" the rendering because the produced state changes its receivers and its next transition.

The complete operation can now be named **Predictive Neural Rendering**: state-conditioned, recurrent construction of a situated internal scene from learned sensory, mnemonic, affective, bodily, and action-related relations, where each constructed state constrains and updates the next.

5. A state-conditioned formal model

Let x_t denote the distributed latent scene at time t. It includes object, spatial, bodily, affective, and action-related components. Let m_t denote reactivated memory material, e_t current external evidence, b_t body-state evidence, and s_t the sleep and neuromodulatory state. These are model variables, not claims that one instrument can observe the entire scene. A minimal transition model is

$$x_{t+1} = F_\theta(x_t, m_t, e_t, b_t; s_t) + \epsilon_t. \tag{1}$$

The parameter set θ represents learned structure. The noise term ϵ_t includes unmodeled variation but does not imply that dreams are unconstrained randomness.

5.1 State-dependent precision

The system does not weigh every evidence channel equally. Define a state-dependent precision vector

$$\boldsymbol{\pi}_t = (\pi_t^{ext}, \pi_t^{mem}, \pi_t^{body}, \pi_t^{meta}, \pi_t^{motor}). \tag{2}$$

In ordinary waking perception, immediate external evidence often constrains the scene strongly. In dreaming, π_t^{ext} can be reduced or transformed while memory-driven and internally generated constraints retain influence. Lucidity increases metacognitive access without necessarily terminating the scene, so it should not be modeled as a simple return to the waking vector.

The latent scene may be estimated by minimizing a precision-weighted mismatch:

$$\hat{x}_t = \arg\min_x \sum_{k \in K} \pi_t^k \, d_k\big(y_t^k, G_k(x)\big) + \lambda R(x), \tag{3}$$

where $K = \{\mathrm{ext}, \mathrm{mem}, \mathrm{body}, \mathrm{meta}, \mathrm{motor}\}$, G_k predicts a channel-specific observation, d_k is an appropriate mismatch function, and $R(x)$ encodes learned structural constraints. Equation (3) is a modeling proposal. It does not imply that the brain literally evaluates one centralized scalar objective.

5.2 Source attribution

Let $z_t \in \{\mathrm{external}, \mathrm{internal}, \mathrm{mixed}, \mathrm{unknown}\}$ represent the attributed source of a content event. The posterior source classification is

$$P(z_t \mid c_t, s_t, h_t) \propto P(c_t \mid z_t, s_t, h_t) P(z_t \mid s_t, h_t), \tag{4}$$

where c_t is the current content and h_t is recent history. A nonlucid dream can contain vivid internally generated content while assigning high probability to an external dream-world source. Lucidity is modeled as improved classification of the state, not disappearance of the content:

$$L_t = P(z_t = internal \mid c_t, s_t, h_t). \tag{5}$$

Equation (5) makes a discriminating prediction: scene vividness and source classification should sometimes vary independently.

5.3 Memory selection and recombination

Let $M = \{m_1, \ldots, m_n\}$ be candidate memory components. Their state- and context-dependent activation probability is

$$P(m_i \text{ active} \mid x_t, s_t, q_t) = \sigma\big(w_i^\top \phi(x_t, s_t, q_t)\big), \tag{6}$$

where q_t includes experimental cues and σ is a bounded link function. Dream construction is not required to reproduce one stored episode. A recombination index can quantify how much a report or decoded state combines components whose waking sources were separately learned:

$$R_{comb} = 1 - \max_j \frac{|A_{dream} \cap A_j|}{|A_{dream} \cup A_j|}, \tag{7}$$

where A_{dream} is the set of coded dream attributes and A_j is the attribute set of waking episode j. High R_{comb} indicates that no single episode accounts for most of the content. The metric requires preregistered coding and does not prove a neural mechanism by itself.

5.4 Receiver-relative difference

For receiver population r, let $\varphi_r(t)$ be an observed phase or timing state and $\widehat{\varphi}_r(t \mid h_t, s_t)$ its expected relation given history and state. Define a circular phase departure

$$\Delta\varphi_r(t) = \operatorname{Arg}\Big(e^{i[\varphi_r(t) - \widehat{\varphi}_r(t \mid h_t, s_t)]}\Big). \tag{8}$$

A PWD feature vector is broader than this scalar phase term:

$$\mathbf{d}_r(t) = [\Delta\varphi_r, \Delta\tau_r, \Delta f_r, \Delta A_r, \Delta w_r, \Delta\rho_r, \Delta I_r]_t, \tag{9}$$

where the components represent changes in phase, latency, frequency, amplitude, waveform, spatial recruitment, and inhibition/excitation balance. A candidate PWD event requires receiver sensitivity and a downstream consequence:

$$\mathrm{PWD}_r(t) = \mathbb{1}[Q_r(\mathbf{d}_r(t), u_r(t)) > \eta_r]\, \mathbb{1}[\Delta y_{t+1} \neq 0], \tag{10}$$

where $u_r(t)$ is receiver state, Q_r is an estimable detector, η_r is a threshold, and Δy_{t+1} is a measured later effect. This definition distinguishes PWD from a raw phase code, a generic prediction error, or an observer's post hoc description.

5.5 Recurrent reconstruction

Let A_t be an effective directed network among neural populations. A NAPOT-style update is

$$\mathbf{x}_{t+1} = \mathbf{\Psi}_{s_t}(A_t \mathbf{x}_t + \mathbf{m}_t + \mathbf{e}_t + \mathbf{b}_t), \tag{11}$$

where Ψ_{s_t} is a state-dependent receive-transform-project operation. The architecture is distributed: each receiving population transforms a partial relation, and recurrent exchange constrains the next joint state.

Define a scene-consistency term across decoded modalities and transitions:

$$C_t = \sum_{(i,j)\in E} \omega_{ij}\ \mathrm{sim}\Big(D_i(x_t^i), D_j(x_t^j)\Big) - \kappa V(x_{t-1}, x_t), \tag{12}$$

where D_i are modality-specific decoders and V penalizes transitions that violate the currently active scene model. Dreams can tolerate higher waking-world violation while maintaining local consistency because the relevant constraints are those currently active in x_t, not all constraints available to waking cognition.

5.6 Conscious access and later report

Dream generation, dream experience, and dream report must be separated. Let g_t denote scene generation, a_t conscious access, and r_T report after awakening at time T:

$$P(r_T \mid g_{1:T}) = \sum_{a_{1:T}} P(r_T \mid a_{1:T}, s_T, \delta_T) P(a_{1:T} \mid g_{1:T}, s_{1:T}), \tag{13}$$

where δ_T represents report delay and interference. A "no dream" report can result from absent experience, inaccessible experience, or failed later recall. Experiments must therefore include immediate awakenings, confidence, report latency, and, when possible, within-dream signals.

6. What the SAN model adds

Predictive processing already provides top-down generative models and bottom-up residuals [26-28]. Activation-synthesis and related dream theories already explain internally driven sleep activity [25]. Memory research already establishes reactivation, replay, and consolidation [12-18]. Lucid-dream research already separates dream experience from metacognitive recognition [7-11]. The contribution here is not to rename those bodies of work.

SAN joins five distinctions that are often studied separately:

1. **A rendering is a recurrently usable state, not merely decoded content.** The constructed scene must alter later receivers and transitions.
2. **A difference is receiver-relative.** The same event may be informative to one receiver and ordinary to another because their learned states differ.
3. **Source attribution is separable from vividness.** Lucidity can correct the inferred source while the scene remains vivid.
4. **Memory replay is necessary neither for every dream nor for reportable consciousness.** Reactivation can remain outside access, while dreams can recombine rather than replay.
5. **Viewpoint and body state are part of the rendered model.** They are not decorations added after object imagery.

The joined causal proposal is

$$\begin{aligned}&\text{sleep state} \to \text{precision shift} \to \text{memory/body activation} \to \text{receiver transformation} \to \text{PWD}\\ &\quad \to \text{recurrent scene} \to \text{source attribution, affect, and action} \to \text{next scene.}\end{aligned} \tag{14}$$

Every arrow in Equation (14) is a measurement target. The sequence is not validated merely because its ingredients exist.

6.1 Strongest-form model comparison

The paper does not compare Predictive Neural Rendering with weakened summaries of neighboring theories. Each model is represented by the problem it most directly solves and by the measurement that would distinguish its strongest form.

Model family	Strongest problem solved	Variables it foregrounds	What it need not explain alone	Draft 2 discriminator
Activation-synthesis [25]	How internally generated sleep-state activation can be synthesized into dream experience.	Brainstem and forebrain activation, state physiology, synthesis.	Item-specific memory selection, stable source attribution, or receiver-relative differences.	Match sleep state and activation, then test whether structured scene and receiver variables add held-out value.
Generative/predictive processing [26-28]	How hierarchical models generate content under altered precision and update from residuals.	Priors, likelihoods, precision, prediction error, free energy.	A SAN-specific receiver-relative phase/waveform/consequence operator.	Compare capacity-matched predictive-error features with the PWD feature family in Equation (9).
Threat simulation [29]	Why recurrent threatening content could rehearse perception and avoidance.	Threat content, rehearsal, ancestral function.	Nonthreat dreams or the general scene-construction mechanism.	Test threat-specific benefit while controlling for affect intensity, recurrence, and waking rehearsal.
Memory reactivation and elaborative encoding [12-18,31]	How sleep reactivates and recombines recent material in service of later memory.	Replay, cueing, hippocampal-cortical exchange, associative recombination.	Why reactivated material becomes a situated conscious scene or receives a source label.	Dissociate cue-driven memory benefit, reportable incorporation, recombination, and source attribution.
Immersive spatiotemporal hallucination [30]	Why dreams have a situated world, temporal flow, and phenomenal self.	Immersion, viewpoint, self-location, world simulation.	The particular neural receive-transform-project implementation.	Independently perturb viewpoint, object content, source attribution, and transition consistency.
Predictive Neural Rendering	How state, memory, body, viewpoint, source attribution, receiver transformation, and recurrent consequence jointly produce a usable scene.	State-conditioned precision, NAPOT transformation, PWD, scene consistency, source and body models.	It does not claim one universal dream function or rhythm.	Require incremental held-out prediction plus selective perturbational consequences beyond all capacity-matched baselines.

The theories can overlap at the level of ingredients. The empirical question is which variables are necessary for which endpoint, whether their addition improves out-of-sample prediction, and whether a targeted intervention changes the predicted downstream state.

7. Discriminating experiments

7.1 Experiment 1: scene construction versus generic sleep-state classification

Participants learn a controlled virtual environment containing separable objects, locations, agents, body perspectives, and action rules. During subsequent sleep, high-density EEG is combined with scheduled awakenings from REM and NREM. Reports are coded blindly against the learned attribute set.

Three model families are compared:

- **state-only:** predicts dream/no-dream and content only from sleep-stage features;

- **reactivation-only:** adds item-level memory reinstatement;
- **rendering model:** adds cross-modal scene consistency, viewpoint, source attribution, and transition structure.

Held-out performance is measured by

$$\Delta\mathcal{L}_{render} = \mathcal{L}(M_{state}) - \mathcal{L}(M_{render}), \tag{15}$$

with the same participants, feature budget, and regularization. The SAN account gains support only if the rendering variables improve held-out prediction beyond state and item reactivation, and if perturbing a predicted receiver relation selectively changes later content.

7.2 Experiment 2: targeted cueing and recombination

Participants learn two episodes with orthogonal object, location, affect, and action features. During sleep, cues reactivate one feature from each episode. The preregistered primary outcome is cross-episode recombination, not mere cue incorporation. Conditions include no cue, single-episode cue, cross-episode cue, waking incubation, and sham sound.

The causal contrast is

$$\Delta R_{comb} = E[R_{comb} \mid q_{cross}] - E[R_{comb} \mid q_{single/no}], \tag{16}$$

with blinded report scoring and correction for each feature's base rate. If cueing improves memory but does not alter dream content, this separates consolidation from accessible rendering. If dream incorporation changes without later memory benefit, it separates rendered content from consolidation efficacy.

7.3 Experiment 3: source attribution without scene collapse

Lucid dreamers learn a signal for "I know this is internally generated," followed by a vividness rating, viewpoint code, and a predefined dreamed action. The central test asks whether source classification can change while decoded scene and motor-imagery features persist.

Define the dissociation score

$$D_{source} = \Delta L_t - |\Delta V_t|, \tag{17}$$

where L_t is lucidity/source classification and V_t is scene vividness. A positive score indicates improved internal-source attribution with relatively preserved vividness. The result would support a layered model. A simultaneous collapse of scene, vividness, and source classification would favor a less separable architecture.

7.4 Experiment 4: viewpoint and bodily-self perturbation

Before sleep, participants experience first-person and displaced-perspective virtual scenes with matched content. During verified lucid dreams or immediate awakenings, they report viewpoint, self-location, body ownership, and movement intention. Closed-loop cues target one learned perspective. The model predicts that perspective-specific cueing and temporoparietal connectivity will shift self-location more strongly than object identity alone.

The outcome is multivariate:

$$\mathbf{B}_t = [\text{viewpoint}, \text{self-location}, \text{ownership}, \text{agency}, \text{vestibular congruence}]_t. \tag{18}$$

This test links dream phenomenology to experimentally separable bodily-self variables rather than treating "out of body" as an all-or-none category.

7.5 Experiment 5: PWD specificity

For candidate receiver populations, fit four capacity-matched models to predict the next decoded scene transition:

1. power and rate only;
2. raw phase and phase-amplitude coupling;
3. generic predictive error;
4. receiver-relative PWD features from Equation (9).

The PWD model must improve held-out prediction and causal perturbation targeting its estimated receiver window must selectively alter the predicted next transition. Its incremental value is

$$I_{PWD} = I(Y_{t+1}; \mathbf{d}_r \mid X_t, S_t, M_t) - I(Y_{t+1}; B_r \mid X_t, S_t, M_t), \tag{19}$$

where B_r is the strongest baseline feature set. If $I_{\mathbf{PWD}} \leq 0$ under adequate power and calibration, the PWD formalization has not earned its additional vocabulary in that task.

7.6 Experiment 6: computational application proof

A recurrent generative agent learns two partially overlapping virtual worlds. During an offline "sleep" state, external input is attenuated, stored trajectories are reactivated, and the agent generates internal rollouts. The application compares:

- exact replay;
- shuffled replay;
- generative replay without receiver state;
- receiver-conditioned predictive rendering;
- the same model with PWD features ablated;
- the same model with source-attribution and viewpoint heads ablated.

Outcomes include continual-learning retention, recombination accuracy, impossible-transition rate, calibration of internal versus external source, and next-state prediction. A useful model must improve at least one declared outcome without hiding degradation in the others. The application is a mechanistic proof-of-concept, not evidence that the artificial system dreams consciously.

7.7 Acquisition and feasibility contract

Each experiment is bound to an event anchor, acquisition stack, control family, primary endpoint, and declared negative result in the Draft 2 experiment ledger. Experiments 1-5 require new prospective human data because the planned cue identities, within-dream signals, report timing, viewpoint manipulations, and receiver-window perturbations do not coexist in one inherited dataset. Existing studies establish feasibility for the component methods - scheduled awakenings [4], visual-category decoding [5], lucid signaling [7-11,32], targeted cueing [12,13,21], and bodily-self manipulation [22-24] - but they are not silently combined into a synthetic completed experiment.

The biological program uses polysomnographic sleep staging as the minimum state boundary; synchronized EEG/EOG/EMG and stimulus/event logs; prespecified artifact rejection; immediate, time-stamped reports; blinded content coding; participant-level train/test separation; and hierarchical estimates that do not treat repeated reports from one participant as independent people. Experiment 6 begins with generated fixtures so every metric and failure condition can be tested deterministically before any empirical claim. The exact acquisition and analysis commitments are enumerated in `../data/DRAFT2-EXPERIMENT-ACQUISITION-LEDGER.csv`.

7.8 Draft 3 identifiability and parameter-recovery gate

7.8.1 Reconstruction is not identification

A decoder can reproduce measured observations while assigning the wrong meaning to its latent variables. That is the first problem this gate addresses. Index participants by (p), nights by (n), and events by t. Let

$$\boldsymbol{\ell}_{pnt} = [v_{pnt}, z_{pnt}, \mathbf{B}_{pnt}, \mathbf{m}_{pnt}]^\top \tag{20}$$

collect scene vividness (v), source attribution (z), bodily-self state $\mathbf{B}$, and memory/recombination state $\mathbf{m}$. Let $\mathbf{o}_{pnt}^{(q)}$ denote observations from modality q, such as EEG source features, EOG or respiratory signals, cue logs, a prespecified within-dream response, or an immediate structured report. A local observation model is

$$\mathbf{o}_{pnt}^{(q)} = H_{s_{pnt}}^{(q)} \boldsymbol{\ell}_{pnt} + C^{(q)} \mathbf{w}_{pnt} + \mathbf{a}_p^{(q)} + \mathbf{b}_{pn}^{(q)} + \boldsymbol{\varepsilon}_{pnt}^{(q)}, \tag{21}$$

where $\mathbf{w}$ contains measured nuisance variables, $\mathbf{a}_p$ and $\mathbf{b}_{pn}$ represent participant and night variation, and the loading map is permitted to change with sleep state. Equation (21) does not assert that the true biological system is linear. It defines the local surrogate in which recoverability can be checked before a nonlinear model is trusted.

Without anchors, an invertible transformation T gives

$$\boldsymbol{\ell}' = T\boldsymbol{\ell}, \qquad H' = HT^{-1}, \qquad H'\boldsymbol{\ell}' = H\boldsymbol{\ell}. \tag{22}$$

The observations therefore cannot by themselves determine which rotated latent axis is "source," "vividness," or "viewpoint." Flexible nonlinear latent-variable models require still stronger structure; conditional auxiliary variables can restore particular forms of identification, but an unconstrained decoder does not [34]. This paper will not rename an arbitrary learned coordinate as a SAN operation.

7.8.2 Anchor condition for source and vividness

The proposed anchors are operation-specific: verified lucidity signals and post-signal source judgments for (z); independently calibrated vividness reports and content-linked readouts for (v); controlled first-person/displaced-perspective tasks for bodily-self variables; and known cue identities plus blinded attribute coding for memory/recombination variables. After nuisance projection and sign/scale conventions are fixed by those anchors, the weighted information matrix is

$$G_\ell = H^\top W H. \tag{23}$$

Local anchor-identification proposition. In the declared linear-Gaussian surrogate, the anchored latent coordinates have a unique weighted least-squares solution if G_ℓ is positive definite. The solution is

$$\hat{\boldsymbol{\ell}} = (H^\top W H)^{-1} H^\top W \tilde{\mathbf{o}}, \qquad \lambda_{\min}(G_\ell) > 0, \tag{24}$$

where $\tilde{\mathbf{o}}$ is the observation after declared nuisance and hierarchy handling. The proof is immediate: if two solutions produce the same residualized observation, their difference lies in the null space of H; positive definiteness makes that null space trivial. This is a bounded algebraic result, not a proof that any current instrument observes the full dream scene.

For source and vividness alone, separate interpretation requires the two anchored columns h_z and h_v not to collapse onto one direction:

$$\det\begin{pmatrix} h_v^\top W h_v & h_v^\top W h_z \\ h_z^\top W h_v & h_z^\top W h_z \end{pmatrix} > 0. \tag{25}$$

If Equation (25) fails or is numerically unstable, the correct output is not "source equals vividness." It is **source-vividness separation unidentified under this acquisition design**. That refusal protects the claim in Equation (17) from being manufactured by a convenient latent rotation.

7.8.3 Missing reports and source-dependent access

Let $R_{pnt} = 1$ when the event yields an analyzable report or validated within-dream signal. A sensitivity model is

$$P(R_{pnt} = 1) = \sigma\big(\alpha_0 + \boldsymbol{\alpha}^\top \mathbf{w}_{pnt} + \boldsymbol{\xi}^\top \boldsymbol{\ell}_{pnt}\big). \tag{26}$$

When $\boldsymbol{\xi} = 0$ after conditioning on recorded variables, inverse-probability weighting or a joint model can correct a declared missing-at-random mechanism. When an unobserved vividness, source, or recall state directly changes report probability, $\boldsymbol{\xi} \neq 0$, the distribution is not point identified from observed reports alone. Draft 3 therefore requires an MNAR sensitivity grid, not a confident correction. "No report," "no accessible experience," and "no generated scene" remain different outcomes.

7.8.4 Conditional PWD identification

The PWD test must not win merely because it receives more columns than an older baseline. Let B contain the strongest capacity-matched non-PWD predictors available for the same receiver and window: rate, power, raw phase, phase coupling, waveform, state, and generic predictive-error features. Let D contain the candidate receiver-relative departures and receiver-state interactions. Remove the part already predictable from the baseline:

$$P_{B,\lambda} = B(B^\top W B + \lambda I)^{-1} B^\top W, \qquad D_\perp = (I - P_{B,\lambda})D. \tag{27}$$

A local next-transition model is

$$Y_{pnt+1} = B_{pnt}\boldsymbol{\beta} + D_{\perp,pnt}\boldsymbol{\gamma} + a_p + b_{pn} + \epsilon_{pnt}. \tag{28}$$

The PWD parameters are conditionally recoverable in this surrogate only if

$$\lambda_{\min}\left(\frac{1}{N} D_\perp^\top W D_\perp\right) > 0. \tag{29}$$

If the minimum eigenvalue approaches zero, phase, power, waveform, receiver state, and the proposed PWD family cannot be separated under that design. The analysis must refuse a unique $\boldsymbol{\gamma}$, even if a regularized program returns numbers. If Equation (29) holds, PWD still earns support only if the full model improves a preregistered held-out endpoint over both the strongest ordinary baseline and an equal-dimensional nonlinear expansion:

$$\Delta_{PWD} = \min\{\mathcal{L}_{held}(B), \mathcal{L}_{held}(B + A_{matched})\} - \mathcal{L}_{held}(B + D_\perp) > 0. \tag{30}$$

The matched alternative has the same number of added degrees of freedom and the same fitting budget. This makes the comparison about receiver-relative information rather than model size.

7.8.5 Leakage, calibration, hierarchy, and multiplicity

Reports from one participant, one night, or one recording session are not independent people. The outer test split therefore holds out whole participants. Model choice, feature selection, threshold selection, imputation, and normalization occur only inside the training participants. Nights are kept inside participants, and temporal windows belonging to one event cannot cross partitions. Random event-row splitting is reported only as a leakage audit. Cross-validation uncertainty can otherwise be severely understated in neuroimaging [36].

For a binary source estimate $\widehat{p}_i$, Draft 3 reports both discrimination and calibration:

$$\text{Brier} = \frac{1}{N}\sum_i (z_i - \widehat{p}_i)^2, \qquad \text{ECE} = \sum_{b=1}^{K} \frac{|I_b|}{N} \left|\overline{z}_{I_b} - \overline{p}_{I_b}\right|. \tag{31}$$

For the primary PWD contrast, uncertainty is generated by resampling participants first and nights within each selected participant second; event rows remain together at the night level. This matches the nested data rather than shrinking uncertainty by pseudoreplication [35]. A single preregistered primary endpoint carries the confirmatory decision. Secondary content families are labeled exploratory and adjusted as a family; an uncorrected secondary minimum (p)-value cannot rescue a failed primary endpoint.

7.8.6 Deterministic recovery and refusal fixtures

The accompanying program freezes seed 20260813 and generates known-ground-truth nested data. It tests 60 participants for source/vividness recovery, 64 for PWD recovery, and 50 for leakage calibration. Training and test partitions are participant-disjoint. The program uses the same model dimension when comparing event-row and participant-held-out splits, and the PWD alternative is compared with an equal-dimensional nonlinear expansion.

Fixture	Result	Interpretation permitted
Full-rank anchored source/vividness model	loading condition number 1.14; vividness correlation 0.949; source AUC 0.993; source ECE 0.020	Recovery works inside the declared supervised linear surrogate.
Collapsed source/vividness anchors	loading condition number 144,246	Refuse separate source/vividness interpretation despite nonzero predictive scores.
PWD positive fixture	held-out RMSE 0.559 versus strongest matched baseline 0.789; coefficient relative error 4.8%	Recovery works under positive conditional rank and known synthetic truth.
Hierarchical PWD interval	mean squared-error gain 0.317; participant/night bootstrap 95% interval [0.272, 0.361]	The positive fixture survives the declared nested resampling.
Zero-PWD fixture	RMSE gain -0.0006	Correct null: do not report PWD improvement.
Conditionally collinear PWD fixture	minimum conditional eigenvalue 9.94×10^{-11}	Refuse PWD parameter identification.
Leakage fixture	row-split RMSE 0.378; participant-held-out RMSE 0.910	Row splitting is optimistically biased and rejected as primary evidence.
Missingness fixture	MAR weighting absolute bias 0.004; misspecified MAR weighting under MNAR leaves 0.343 bias	MNAR requires sensitivity analysis; it is not repaired by assuming MAR.

All thirteen predeclared fixture checks passed. That means the code recovers declared parameters when the design is identifiable and refuses or returns a null when it is not. It does **not** mean that source attribution, vividness, or PWD has been recovered from human dream data. The complete result, coefficients, uncertainty interval, leakage table, fixture contract, and hashes are in `../application-proof/draft3-identifiability/`.

7.9 Draft 4 executable reference application and confound gate

The computational proposal creates a practical problem before it creates a new term. A system can appear to preserve a world because training and evaluation rows contain the same agent, because replay uses more examples, because a richer model receives more features, because invalid transitions are not counted, or because one favorable accuracy hides failed calibration and recombination. A reproducible test therefore needs typed events, fixed partitions, equal replay budgets, explicit countermodels, simultaneous outcome reporting, active refusals, deterministic replay, and a final partition that the development run cannot inspect.

Draft 4 implements that contract in `../application-proof/draft4-reference-application/`. Its environment is deliberately transparent rather than photorealistic. Eighteen synthetic agents learn two categorical worlds. Eight disjoint agents supply development events from World A, World B, and a hybrid world that recombines components learned separately. This makes the generating rule known and lets the software be tested without presenting its output as dream physiology.

7.9.1 Typed events and protected partitions

Each transition is serialized as a versioned record

$$\mathcal{E}_i = (a_i, w_i, e_i, t_i, s_i, c_i, u_i, d_i, o_i, \ell_i, v_i, z_i, \alpha_i, \mathbf{b}_i, y_i, \mathcal{A}_i), \tag{32}$$

where a is agent identity, w world, e episode, t step, s split and state, c memory cue, u receiver state, d PWD bin, (o, ℓ, v, z) current object, location, viewpoint, and source, α action, $\mathbf{b}$ ordinary rate, power, phase, coupling, waveform, and prediction-error bins, y the target transition, and $\mathcal{A}$ the set of allowed transitions. Runtime types and a JSON schema reject missing fields, unknown enumerations, duplicated identifiers, incomplete training targets, and any target attached to the future final-test split.

The fixed fixture contains 864 World A training transitions, 864 World B training transitions, 480 development transitions, and 96 future feature-only transitions. Training and development agents do not overlap. The future records contain no target or allowed-transition labels.

7.9.2 Sequential learning and the twelve-arm comparison

For arm m, a model first learns World A. It then learns World B together with an arm-specific replay set drawn under one fixed budget:

$$\hat{\theta}_m = operatornameFit_m(D_B \cup R_m(D_A)), \qquad |R_m(D_A)| = 432. \tag{33}$$

The twelve arms preserve every countermodel named in Sections 6.1, 7.5, and 7.6: state only; item reactivation only; power and rate; raw phase and phase-amplitude coupling; generic predictive error; an ordinary-signal comparator with two nonlinear interactions matching the receiver/PWD feature count; exact replay; shuffled replay; generative replay without receiver state; receiver-conditioned predictive rendering; PWD ablation; and source/viewpoint-head ablation. All arms use the same categorical learner, smoothing rule, World B set, replay count, and development partition. The comparison therefore does not grant the proposed arm extra trials or a private evaluation endpoint.

The replay operator is explicitly part of the manipulation:

$$R_m \in \{R_{exact}, R_{shuffled}, R_{generated\setminus receiver}, R_{receiver}, R_{receiver\setminus PWD}, R_{receiver\setminus source,viewpoint}\}. \tag{34}$$

The state, reactivation, electrophysiology, and predictive-error feature families receive exact replay so that their failure cannot be attributed simply to corrupted stored transitions.

7.9.3 One metric cannot conceal another

Every arm emits the complete vector

$$\mathbf{V}_m = [A_{next}, A_{component}, A_{retain}, A_{recombine}, R_{impossible}, B_{source}, ECE_{source}, A_{viewpoint}]_m. \tag{35}$$

The development results are:

Arm	Next	Component	Retention A	Hybrid	Impossible	Brier	ECE	Viewpoint
state only	0.004	0.414	0.000	0.000	0.988	0.042	0.007	0.508
reactivation only	0.017	0.487	0.013	0.013	0.960	0.043	0.006	0.533
power and rate only	0.004	0.414	0.000	0.000	0.988	0.042	0.007	0.508
raw phase and coupling	0.010	0.416	0.000	0.006	0.977	0.042	0.013	0.527

Arm	Next	Component	Retention A	Hybrid	Impossible	Brier	ECE	Viewpoint
generic predictive error	0.123	0.647	0.175	0.019	0.769	0.043	0.006	0.542
capacity-matched ordinary	0.196	0.733	0.250	0.056	0.629	0.045	0.088	0.881
exact replay	0.775	0.923	0.988	0.338	0.225	0.015	0.115	1.000
shuffled replay	0.340	0.684	0.000	0.081	0.650	0.102	0.146	0.825
generated replay without receiver state	0.202	0.734	0.263	0.050	0.631	0.041	0.061	0.892
receiver-conditioned predictive rendering	0.775	0.923	0.988	0.338	0.225	0.015	0.115	1.000
PWD ablated	0.281	0.777	0.406	0.056	0.575	0.050	0.079	1.000
source/viewpoint heads ablated	0.373	0.736	0.531	0.031	0.627	0.250	0.217	0.533

The receiver-conditioned arm improves exact next-state accuracy by 0.579 over the capacity-matched ordinary comparator and by 0.573 over generated replay without receiver state. PWD ablation reduces exact accuracy to 0.281 in the positive fixture. Removing source and viewpoint heads exposes the expected head-specific losses. Shuffling replay destroys World A retention while leaving World B performance high, demonstrating why one aggregate score would be misleading.

The adverse results are equally important. Receiver-conditioned generated replay does not outperform exact replay; the two arms tie on every reported development measure. Hybrid exact accuracy is only 0.338. The proposed arm still produces an impossible transition on 0.225 of development events. Its source Brier score is low, but ECE is 0.115 and exceeds the application's 0.10 diagnostic calibration limit. Draft 4 therefore establishes executable separation from weaker baselines under the planted rules, while leaving constrained decoding, hybrid generalization, and training-only recalibration open. Those weaknesses were not tuned away after inspecting development data.

7.9.4 Null, leakage, and refusal controls

The PWD-null fixture regenerates the targets so that PWD has no causal role. Receiver-conditioned and PWD-ablated arms both reach 0.754 exact accuracy, producing an apparent PWD gain of exactly 0.000. This is the required null, not a failed favorable run.

The leakage audit gives the same agent-specific label to repeated rows. Random row splitting reaches 1.000 accuracy because each agent appears on both sides; holding out entire agents yields 0.375, an optimism gap of 0.625. The primary application never uses the row split.

Ten executable refusal tests reject: a missing required field; a duplicate event identifier; target leakage into the future partition; agent overlap between training and development; an opened future seal; an unknown enumeration; an artificial-consciousness inference; deterministic aliasing of PWD with raw phase; materially miscalibrated source probabilities; and a tampered replay record. Refusal is a successful outcome when the design cannot support the requested interpretation.

7.9.5 Deterministic replay and the unopened final test

Every prediction is written into a hash chain

$$h_i = \mathrm{SHA\,256}(i, m_i, e_i, \hat{y}_i, h_{i-1}), \qquad h_0 = 0^{64}. \tag{36}$$

All 5,760 records verify. Two independent executions produce the same complete replay hash, `983835C54F8AB3130AD3A1DDF292A057E7948ACF7297F4790794BFAFA54AAE26`. A one-field tamper breaks verification.

The future partition is defined by

$$D_{future}^{features} \neq \varnothing, \qquad D_{future}^{targets} = \varnothing, \qquad \text{status}(D_{future}) = \text{CLOSED}. \tag{37}$$

Its 96 feature records are bound by SHA-256 commitment `ECEF31A49B3CD3B2DCD5F75213F41E56F475B271D934D03B465DACBFAECE8E04`. Because labels are absent, Draft 4 cannot calculate a final-test metric. A later gate must freeze its training-only repairs, create a new opening receipt, and leave the original development outcomes unchanged.

This is a procedural seal, not a cryptographic substitute for external custody. The categorical generator is transparent and local, so a determined analyst could reconstruct its rule. A later empirical final test must be acquired or held by an independent process that the development code cannot reproduce.

7.9.6 What this application does and does not establish

The application establishes that the paper's computational comparison is executable, typed, reproducible, leakage-aware, countermodel-complete at the declared categorical scale, and capable of returning favorable, null, adverse, and refusal outcomes. It shows that receiver and PWD variables matter when the transparent generator actually uses them. It does not show that biological dreams use the same rule, that PWD has been measured in a sleeping brain, that the categorical learner implements NAPOT, or that the agent has an experience. Those require the prospective human and higher-capacity computational gates already specified.

7.10 Draft 5 development repair and generalization gate

Draft 4 exposed three implementation problems before a final test could be justified. Independently predicted object, location, viewpoint, and source components could be combined into a transition that the world never permits. A probability score could discriminate internal from external source while remaining poorly calibrated. A learner could remember each training world yet fail when valid components were recombined in a new world. These are ordinary modeling failures, not new SAN vocabulary. Draft 5 therefore asks whether training-only constraints, training-only calibration, and compositional baselines repair them across multiple fresh development fixtures without opening the future partition.

The complete repair was frozen before outcomes. Four of eighteen training agents were reserved for calibration; the other fourteen supplied model fitting. Ten fresh deterministic seeds each generated a new 480-event development set. Draft 4's development seed was excluded. No development target selected a temperature, intercept, candidate transition, arm, or threshold.

7.10.1 Constrain a prediction before naming it valid

The repair first learns which complete transitions can be assembled from the fit-training events. For an event e, the unconstrained learner assigns scores $s_\theta(y \mid e)$ to possible next states, while a separately learned training-only operator supplies a small admissible candidate set $\widehat{\mathcal{A}}_{train}(e)$. The constrained decoder returns

$$\hat{y}(e) = \underset{y \in \widehat{\mathcal{A}}_{train}(e)}{\operatorname{argmax}} \; s_\theta(y \mid e). \tag{38}$$

This does not declare the learned set biologically correct. It tests a narrower engineering proposition: a model should not receive credit for a high component score while emitting a combination that its own learned environment treats as impossible.

7.10.2 Calibrate source estimates without looking at development outcomes

Calibration-only agents supply probabilities p_i and source labels z_i; the training-only minimum-NLL choices are:

$$\tilde{p}_i = \sigma\left(\frac{\text{logit}(p_i) + b}{T}\right), \qquad (\widehat{T}, \hat{b}) = \underset{T,b}{\operatorname{argmin}} \left[-\sum_i z_i \log \tilde{p}_i + (1 - z_i) \log(1 - \tilde{p}_i)\right]. \tag{39}$$

The frozen grid contained nine temperatures and five intercepts. All ten primary runs selected $T = 0.25$ and $b = 0$ from 384 calibration events. Development-set expected calibration error was reported afterward; it did not tune the transformation.

7.10.3 Stronger baselines and multi-seed results

Eleven arms were evaluated on every seed. In addition to repaired versions of the Draft 4 learners, the gate added receiver-aware and ordinary-feature factorized transition models. These models learn object shifts, location shifts, viewpoint changes, and source changes as separately reusable training relations. They are a stronger test than replay alone because the hybrid world recombines familiar relations in configurations absent from either training world.

Arm	Median exact	Minimum exact	Median hybrid	Median impossible	Median source ECE
receiver-conditioned, constrained, calibrated	0.915	0.892	0.756	0.000	0.00078
receiver-conditioned, unconstrained, calibrated	0.743	0.717	0.244	0.257	0.00078
exact replay, calibrated	0.740	0.700	0.241	0.257	0.00080
capacity-matched ordinary, calibrated	0.188	0.158	0.050	0.630	0.06858
generated replay without receiver state, calibrated	0.192	0.158	0.047	0.651	0.06409
PWD-ablated, constrained, calibrated	0.268	0.225	0.206	0.521	0.07179
factorized receiver-aware, calibrated	**1.000**	**1.000**	**1.000**	**0.000**	**<0.000000001**
factorized ordinary-feature, calibrated	0.260	0.235	0.200	0.529	0.01187

The predeclared primary arm passed its four implementation thresholds: zero median impossible transitions, source ECE below 0.10, median hybrid accuracy above 0.50, and exact accuracy above 0.60 on every seed. Constraining the receiver-conditioned learner increased median exact accuracy by 0.172 over its calibrated unconstrained counterpart. Across seeds, the primary exact accuracy ranged from 0.892 to 0.929 and hybrid accuracy from 0.713 to 0.788.

The strongest result, however, belongs to the factorized receiver-aware baseline. It recovered every development transition on every seed. The proposed primary arm therefore does **not** win the strongest comparison. The transparent generator is almost exactly described by the factorization that this compact model learns. Draft 5 consequently establishes that the engineering repairs work inside this categorical benchmark, while also showing that the benchmark is too structurally aligned and too easy to adjudicate the necessity of the larger replay architecture.

7.10.4 Compute and storage are part of the comparison

For each arm and seed, the application records fit and calibration events, replay count and serialized bytes, learned keys and count entries, serialized model and constraint bytes, total stored bytes, and candidate evaluations per development event. Let

$$C_m = (N_{fit}, N_{cal}, N_{replay}, B_{replay}, K_{table}, N_{count}, B_{model}, B_{constraint}, N_{candidate})_m \tag{40}$$

denote that resource vector. The receiver-conditioned constrained arm used 17,390--17,638 learned keys and approximately 2.47--2.50 MB of serialized model, constraint, and replay state, evaluating two candidate transitions per development event. The factorized receiver-aware model used 60 learned keys, approximately 3.9 KB, no replay, and one candidate evaluation. Its perfect score with far lower storage is an adverse simplicity result for any claim that the larger architecture is required by this benchmark.

7.10.5 The null and leakage controls remain decisive

The PWD-invariant control did not pass. Across ten null fixtures, the apparent gain assigned to the full arm ranged from 0.263 to 0.317, with median 0.299, exceeding the frozen absolute limit of 0.03 on every seed:

$$\Delta^{null}_{PWD} = A^{null}_{full} - A^{null}_{ablated}, \qquad \text{median}(\Delta^{null}_{PWD}) = 0.299. \tag{41}$$

This is a spurious effect, not support for PWD. The implementation audit explains why: the nominally full and PWD-ablated arms also use different candidate-set learners, one receiver-aware and one ordinary-feature based. The contrast therefore changes more than PWD. Draft 5 does not license a PWD-specific interpretation; a later repair must hold the constraint learner and every non-PWD feature fixed while removing only PWD information.

The row-split control again showed severe optimism. Across ten fixtures, random row splitting exceeded held-out-agent accuracy by 0.625--0.875, with median 0.750:

$$\Delta_{leak} = A_{random\ row} - A_{held\ agent}, \qquad \text{median}(\Delta_{leak}) = 0.750. \tag{42}$$

Whole-agent separation remains mandatory. This adverse control is preserved alongside the favorable development metrics rather than averaged away.

7.10.6 Reproducibility, retained failures, and the honest gate

The main evaluation contains 110 arm-by-seed rows and 52,800 arm-event evaluations, plus ten PWD-null controls and ten row-split audits. Two complete executions produced byte-identical copies of all ten output files. The result package retains the Draft 4 exact-replay tie, weak hybrid accuracy, impossible transitions, source-calibration failure, PWD null, and row-split optimism, then adds every Draft 5 seed-level null and leakage result. An initial interactive invocation reached the environment's command-return timeout after writing a complete package; the full replay was rerun with an adequate bounded runtime and verified byte for byte. No scientific outcome changed.

The future partition remains closed. Draft 5 read only its seal metadata and never opened the blinded feature file. Because the factorized receiver-aware model dominates and the nominal PWD ablation fails its null, the next honest gate is **not** to open that partition. It is to freeze a harder, independently generated development family; match the strongest factorized baseline; create a one-variable PWD ablation with an identical constraint learner; and demonstrate that the revised null returns no spurious gain. Only after those conditions pass should an independently custodied final test be considered.

7.11 Draft 6 harder-family development-evidence gate

The word *draft* describes the manuscript version. It does not count independent evidence. The scientific object tested here is the **D6 harder-family development gate**: a synthetic model-development comparison performed before any final-test opening. That distinction prevents a sixth manuscript from sounding like a sixth experiment or a final evidentiary stage.

7.11.1 A generator specified before the models were scored

The Draft 5 generator could be solved perfectly by sixty factorized keys. Draft 6 therefore freezes a new generator before writing the model contract or observing an outcome. Its four families are:

1. **cross-coupled:** receiver-by-PWD and phase-by-power interactions alter several target components;
2. **role reversal:** the signed PWD contribution reverses with receiver state and interacts with cue and waveform state;
3. **sequence hysteresis:** exposed step history, current source, current viewpoint, receiver state, and PWD jointly affect the next relation; and

4. **ambiguous mixture:** two admissible rules share the same measured context, so exact accuracy need not reach one.

For family f, observed context x, PWD value p, and declared hidden branch z, the target relation is generated as

$$y \sim G_f(x, p, z; \eta_f), \qquad y_{null} \sim G_f(x, 0, z; \eta_f), \tag{43}$$

where the null holds the random stream, event identity, observed context, and hidden branch fixed while setting only the target-rule PWD term to zero. Eight frozen seeds each produce 1,280 World A training events, 1,280 World B training events, and 1,440 development events from ten disjoint agents across World A, World B, and a hybrid world.

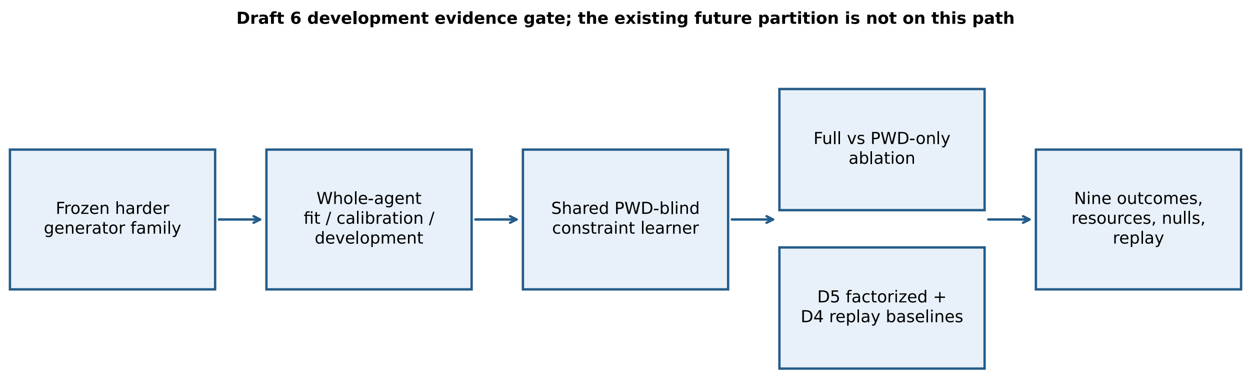

Figure 1. Draft 6 evidence path. The existing future partition is deliberately absent from the execution path.

7.11.2 A PWD-only ablation

Draft 5's PWD comparison also changed its candidate-set learner. Draft 6 removes that confound. Full and ablated arms use the same rows, relational model class, calibration partition, non-PWD variables, and exact same in-memory candidate learner. That shared learner is blind to PWD. The comparison is

$$\hat{y}_{full} = M(X_{nonPWD}, p; C), \qquad \hat{y}_{-PWD} = M(X_{nonPWD}; C), \tag{44}$$

with one shared constraint object C. All eight run receipts verify the same constraint identity, an identical non-PWD feature contract, and `pwd_bin` as the only declared feature difference.

The retained Draft 5 factorized receiver model is not weakened or replaced. Its architecture still learns four separate object-shift, location-shift, viewpoint, and source tables. Draft 6 fits that exact architecture to the new training family and measures it again. Draft 4 exact replay and capacity-matched ordinary models, an ordinary relational model, an unconstrained relational model, and a shuffled-target negative control remain visible as additional comparators.

7.11.3 The harder family breaks the easy result

Arm	Median exact	Minimum exact	Median hybrid	Median impossible	Median source ECE
relational full, shared constraint	**0.128**	0.122	**0.132**	0.831	0.144
relational PWD ablated, same constraint	0.092	0.079	0.098	0.877	0.155
relational receiver and PWD ablated	0.090	0.082	0.092	0.882	0.152

Arm	Median exact	Minimum exact	Median hybrid	Median impossible	Median source ECE
relational full, unconstrained	0.134	0.124	0.139	0.832	0.035
retained D5 factorized receiver	0.040	0.032	0.040	0.939	0.054
retained D5 factorized ordinary	0.035	0.028	0.036	0.946	0.042
D4 exact replay	0.016	0.011	0.004	0.973	0.038
D4 capacity-matched ordinary	0.015	0.012	0.002	0.974	0.044
relational shuffled-target negative	0.082	0.071	0.078	0.890	0.142

The harder generator passes its anti-saturation purpose: the retained factorized receiver architecture never exceeds 0.046 exact accuracy on any seed. The relational primary exceeds its median by 0.089. That relative result does not rescue the primary. It fails three frozen absolute gates: minimum exact accuracy is 0.122 rather than at least 0.35; median hybrid accuracy is 0.132 rather than at least 0.30; and median impossible-transition rate is 0.831 rather than at most 0.35. Only the median ECE limit passes.

The learned constraint is itself inadequate. It is PWD-blind and shared correctly, but its non-PWD context keys admit relations that are not valid for the event's exact hidden branch and family state. Constrained decoding therefore does not guarantee event-specific validity merely because it chooses from a learned candidate set.

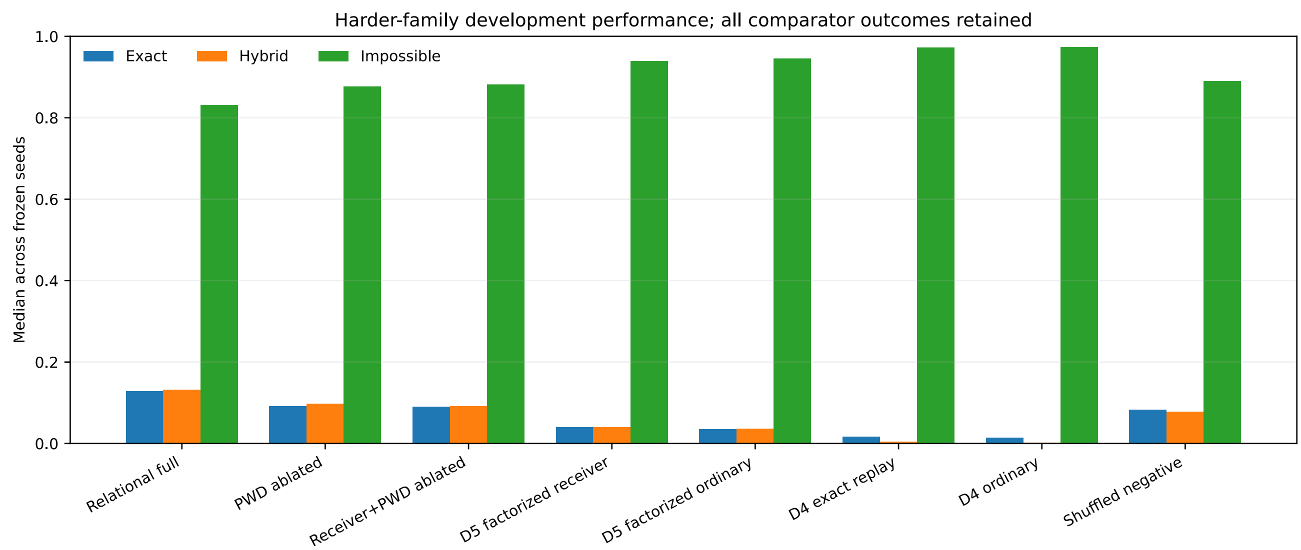

Figure 2. Harder-family development outcomes. Relative improvement coexists with poor absolute performance and high invalid-transition rates.

Family-level results locate the failure instead of averaging it away:

Generator family	Primary median exact	Seed range
F1 cross-coupled	0.081	0.075-0.125
F2 role reversal	0.057	0.042-0.064
F3 sequence hysteresis	0.325	0.303-0.339
F4 ambiguous mixture	0.047	0.031-0.061

The current relational hierarchy handles the exposed sequence family substantially better than the cross-coupled, sign-reversing, or ambiguously mixed families. A later repair must address those named structures rather than tune only the aggregate score.

7.11.4 Positive difference and the true null

On the positive generator, the paired exact-accuracy difference is

$$\Delta^{+}_{PWD} = A^{+}_{full} - A^{+}_{-PWD}, \qquad \text{median}(\Delta^{+}_{PWD}) = 0.0378, \tag{45}$$

with seed range 0.0285-0.0438. The exact sign-permutation diagnostic is $p = 0.0078125$. This is a deterministic multi-seed diagnostic, not a human-sample significance claim.

The true null is much closer to zero than Draft 5's confounded 0.299 result, but it does not pass the frozen maximum bound:

$$\text{median}(\Delta^{null}_{PWD}) = -0.0163, \qquad \Delta^{null}_{PWD} \in [-0.0354, -0.00694]. \tag{46}$$

One or more seeds exceed the absolute 0.03 limit, and the sign-permutation diagnostic again yields $p = 0.0078125$. The negative direction means that the additional PWD dimension systematically hurts this model under the true null, probably through sparse conditional estimation. It is still a specificity failure. Draft 6 therefore does not interpret the positive difference as evidence for biological PWD.

Random row splitting also retains its known failure: the optimism gap ranges from 0.625 to 0.875, median 0.750. Whole-agent holdout remains mandatory.

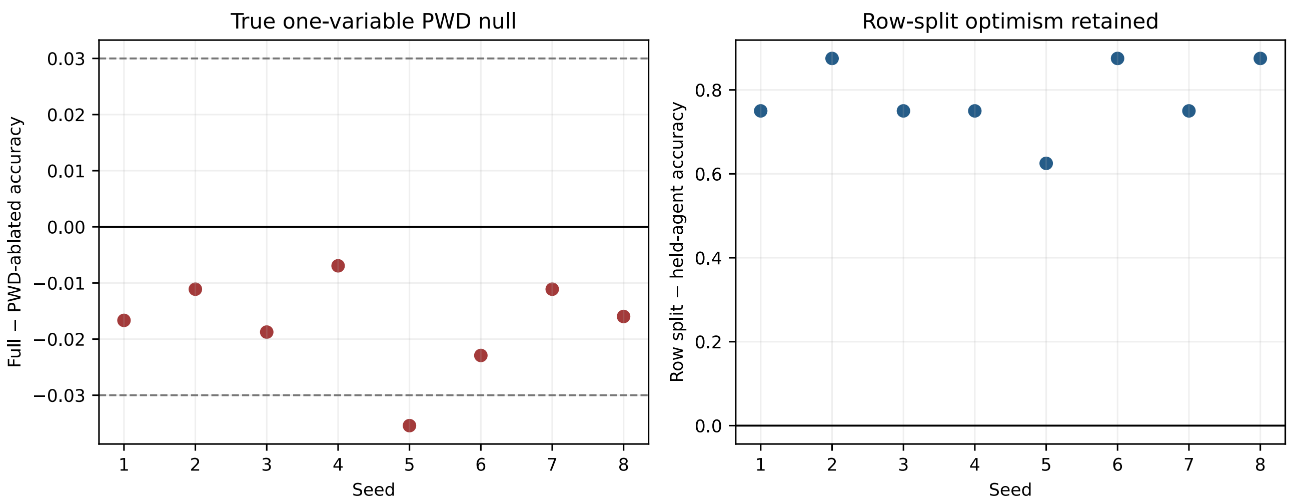

Figure 3. Controls that govern interpretation. The null is improved but not inside its full frozen bound; row leakage remains large.

7.11.5 Resource accounting and exact replay

The same sixty-key factorized receiver architecture grows from 3.9 KB in Draft 5 to 5.24 KB in Draft 6 because its stored counts change, while its key count stays fixed. Across seeds, the relational primary uses 5,606-5,684 learned model keys plus a 599,416-604,611-byte shared constraint and totals approximately 1.69-1.70 MB. Resource efficiency can be summarized by

$$\rho_m = \frac{A_{exact,m}}{B_{model,m} + B_{constraint,m} + B_{replay,m}}, \tag{47}$$

but no single efficiency ratio replaces the full accuracy, hybrid, validity, calibration, family, and null vectors. The smaller factorized model remains an essential baseline even when its absolute accuracy falls.

The main evaluation contains 72 arm-by-seed rows and 103,680 arm-event evaluations, plus eight true-null fixtures and eight row-split audits. Two executions produce fifteen byte-identical output files, including three deterministic SVG figures. The new adverse ledger begins with all 28 Draft 5 rows in their original order, thereby retaining every recorded Draft 4 and Draft 5 adverse, null, and leakage result before adding Draft 6 outcomes.

7.11.6 Honest stopping condition

Draft 6 closes the requested harder-family gate, but it does not clear it for final testing. The next gate must be another **development** repair. It must improve event-specific admissibility without using hidden labels, raise absolute performance on F1, F2, and F4, preserve the retained factorized baseline, and bring every one-variable PWD-null seed inside the frozen bound. The existing future partition remains closed and should not be opened for this model family.

7.12 Draft 7 registered null calibration and estimator intervention

Draft 6 left two separable problems. Its PWD-only contrast was correctly matched but crossed the strict null bound on one seed, while its relational estimator still failed the absolute accuracy and transition-validity criteria. Draft 7 therefore asks two ordered questions. First, can the strength of the PWD contribution be selected without looking at a positive outcome? Second, after that intervention is frozen, does a richer evidence estimator recover the planted positive relation while remaining null-safe? Manuscript Draft 7 documents this one development gate; it is not a seventh independent experiment or a final test.

7.12.1 Three disjoint development partitions

The Draft 6 harder generator remains unchanged. Before Draft 7 outcomes, three new, pairwise-disjoint seed families were registered:

1. six PWD-null fixtures used only to select the intervention multiplier;
2. eight positive fixtures used only after selection to measure the planted effect and all performance endpoints; and
3. eight independent PWD-null fixtures used only after selection to evaluate specificity.

Every fixture contains 2,560 training events and 1,440 held-agent development events. None uses the existing future partition, whose seal remains closed. This is internal development separation, not external custody.

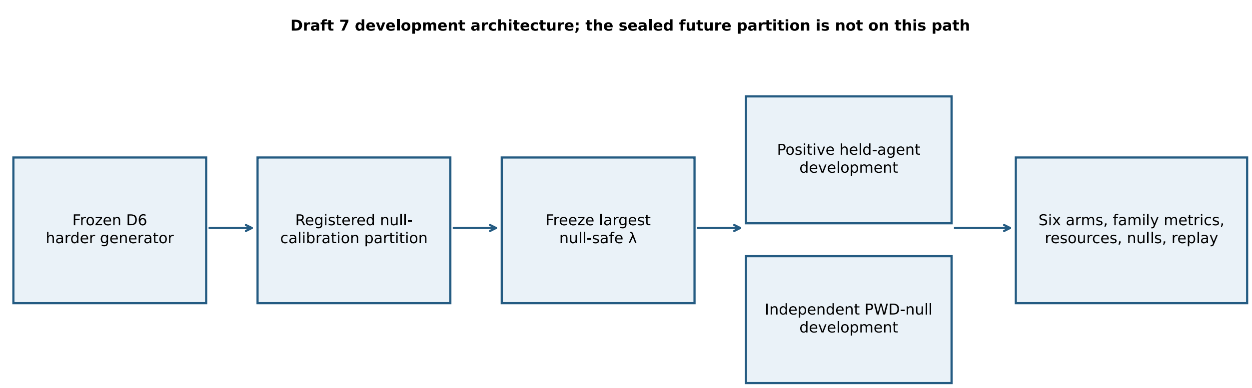

Figure 4. Draft 7 development architecture. The existing future partition is not on the execution path.

7.12.2 A separable evidence estimator

The Draft 7 estimator scores each candidate joint transition relation r from two additive evidence families. The base family contains 211 non-PWD singleton, pair, and condition-centered interaction terms. The PWD family contains 43 terms linking PWD to receiver state, condition, and other measured context. For event e, the registered score is

$$S_\lambda(r \mid e) = S_0(r \mid e) + \lambda S_{PWD}(r \mid e), \tag{48}$$

where each observed term contributes a length-weighted, Laplace-smoothed log count. A twelve-relation shortlist is selected from S_0 alone:

$$\mathcal{C}_{12}(e) = \mathrm{Top}_{12}\{S_0(r \mid e) : r \in \mathcal{R}\}. \tag{49}$$

Full and ablated predictions share one fitted estimator object, one PWD-blind shortlist, one source-probability calibrator fitted on calibration-only training agents, the same rows, and the same non-PWD terms. Their only intervention difference is

$$\hat{r}_{full} = \arg\max_{r \in \mathcal{C}_{12}(e)} S_{\widehat{\lambda}}(r \mid e), \qquad \hat{r}_{-PWD} = \arg\max_{r \in \mathcal{C}_{12}(e)} S_0(r \mid e). \tag{50}$$

Neither target labels nor `allowed_targets` are read during prediction.

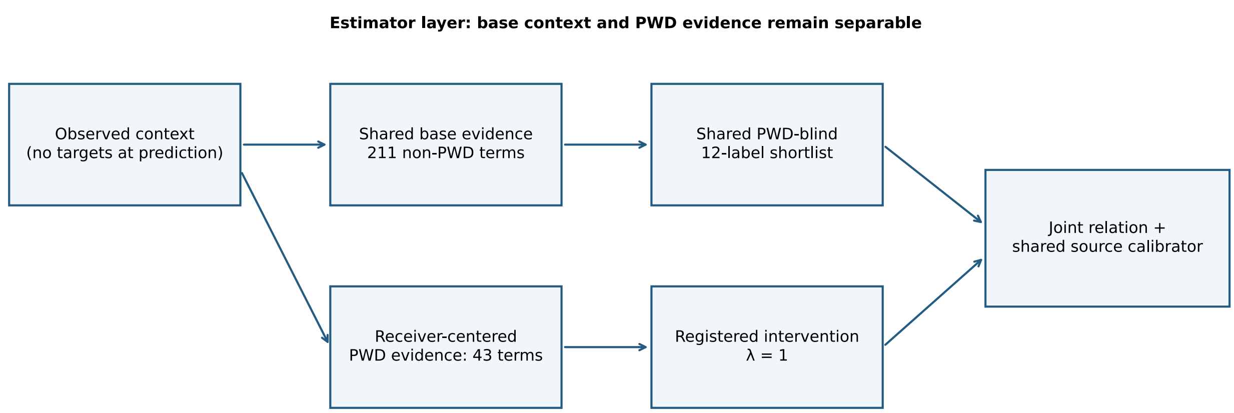

Figure 5. Draft 7 estimator. Separability makes the intervention measurable; it does not guarantee that the architecture is adequate.

7.12.3 The multiplier is selected on nulls, not positive outcomes

The candidate grid was frozen as

$$\Lambda = \{0, 0.05, 0.10, 0.20, 0.35, 0.50, 0.75, 1.00\}. \tag{51}$$

For every candidate, all six null-calibration fixtures report exact and hybrid full-minus-ablated differences. The registered rule selects the largest multiplier for which every seed remains within 0.015 on both endpoints and the median absolute differences remain within 0.0075:

$$\widehat{\lambda} = \max\left\{\lambda \in \Lambda : \max_j |\Delta_{j,\lambda}^{null,exact}| \le 0.015, \max_j |\Delta_{j,\lambda}^{null,hybrid}| \le 0.015, \mathrm{med}_j\, |\Delta_{j,\lambda}| \le 0.0075\right\}. \tag{52}$$

All candidates pass. The rule therefore selects $\widehat{\lambda} = 1.0$, whose worst absolute calibration-seed differences are 0.00694 for exact accuracy and 0.00833 for hybrid accuracy. This is a genuine registered selection result, but it does not rescue an estimator merely by approving its full PWD weight.

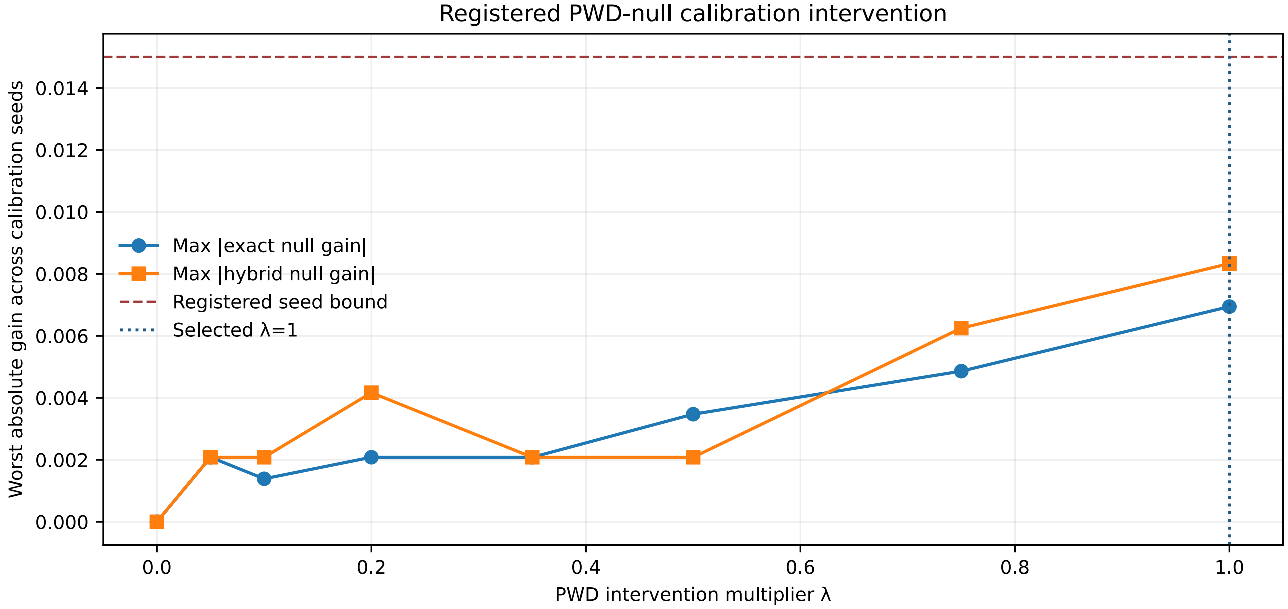

Figure 6. Registered null-calibration intervention. The full multiplier is null-safe inside the calibration partition.

7.12.4 The independent null passes, but the positive mechanism is not recovered

The selected multiplier also passes the eight independent null fixtures. Its exact difference has median -0.00139 and range -0.00694 to 0.000694; its hybrid difference has median -0.00312 and range -0.01042 to 0.00417. Every value lies within the independently frozen 0.02 absolute bound.

On the positive fixtures, however,

$$\mathrm{median}(A_{full} - A_{-PWD}) = 0.00278, \qquad \Delta^{+}_{PWD} \in [-0.00556, 0.00556]. \tag{53}$$

This fails the registered 0.02 median positive-gain floor. The appropriate conclusion is not that PWD has been disproved. The planted generator is known to use PWD, while this estimator fails to recover it. The result therefore identifies a failed estimator architecture at this endpoint.

7.12.5 Performance, families, and resources

Arm	Median exact	Minimum exact	Median hybrid	Median impossible	Median source ECE
D7 null-calibrated full	**0.0358**	0.0264	0.0313	0.9462	0.0647
D7 PWD ablated	0.0326	0.0222	0.0333	0.9507	0.0577
D7 full multiplier 1.0	0.0358	0.0264	0.0313	0.9462	0.0647
D7 shuffled-target negative	0.0167	0.0139	0.0198	0.9667	0.1251
retained D6 relational	**0.1306**	0.1174	**0.1385**	0.8313	0.1289
retained D5 factorized receiver	0.0396	0.0285	0.0344	0.9389	0.0514

The Draft 7 primary fails the minimum exact, median exact, hybrid, impossible-transition, baseline-gain, positive-PWD, and all family floors. Its family medians are 0.0403 for F1, 0.0222 for F2, 0.0597 for F3, and 0.0222 for F4. It passes source calibration, the registered null-calibration rule, and the independent null.

The result is also resource-adverse. Across seeds, the shared Draft 7 estimator contains 5,202 learned keys and an estimated 6.55-6.65 MB of table material. The retained Draft 6 relational model reaches much higher accuracy with approximately 1.69-1.70 MB including its constraint, while the sixty-key factorized baseline uses about 5.24 KB.

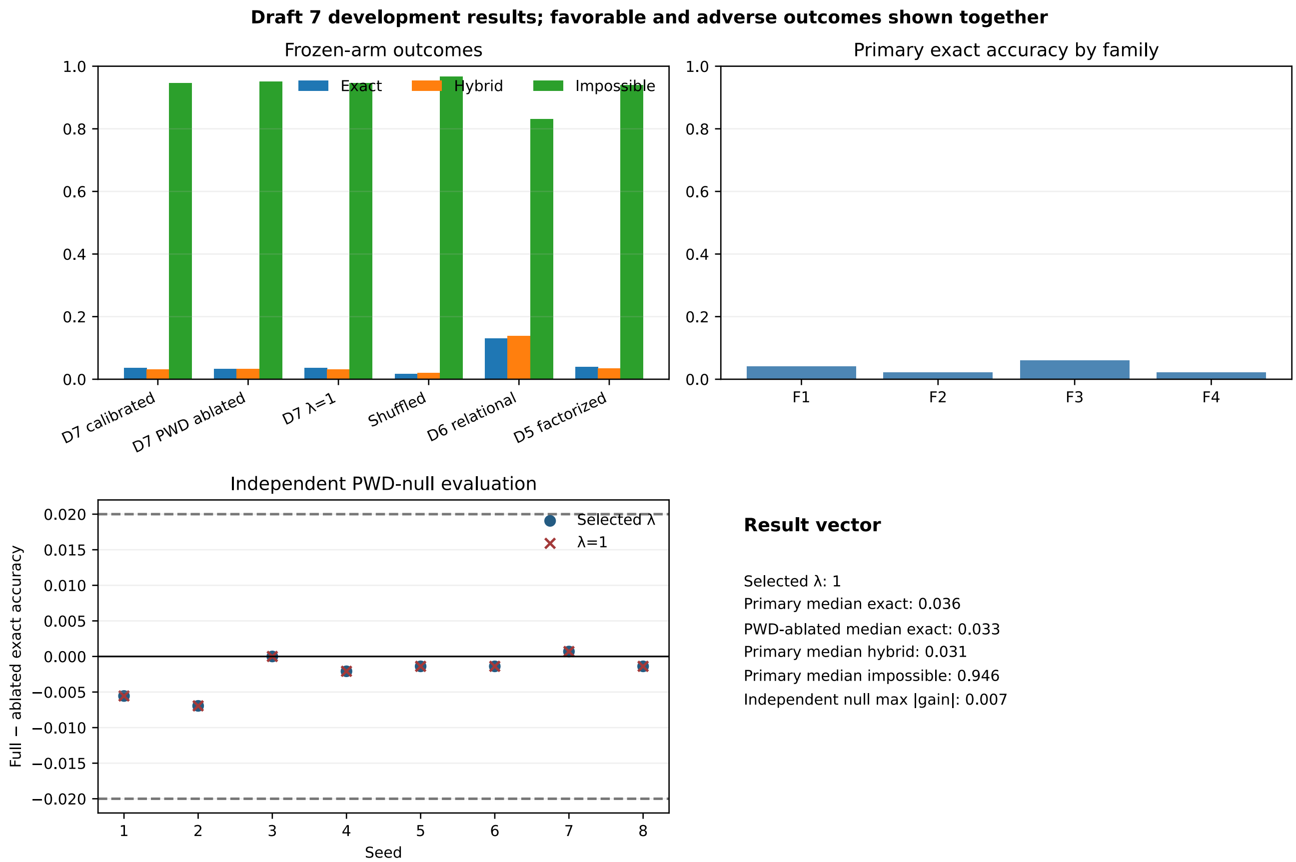

Figure 7. Draft 7 result vector. Passing null specificity and source calibration does not override failed positive recovery, prediction, transition validity, or efficiency.

7.12.6 Exact replay, adverse retention, and stopping rule

The positive evaluation contains 48 arm-by-seed rows and 69,120 arm-event evaluations. The null-calibration ledger contains all 48 seed-by-multiplier rows. Two executions produce nineteen byte-identical output files, including all four SVG figures. The Draft 7 adverse ledger begins with all 49 Draft 6 rows in their original order, then retains every null-calibration candidate, every failed gate, every independent-null seed, and every leakage seed. Row-split optimism remains adverse, with median 0.750 and range 0.500-1.000.

Draft 7 closes the registered null-calibration question for this estimator: full PWD weight is null-safe here. It does not clear the evidence program. A subsequent development gate must replace the failed evidence estimator while keeping the registered partition logic, the same strong baselines, explicit family reporting, resource accounting, and a fresh independent null. The existing future partition remains closed.

8. Falsifiers and failure conditions

The paper's distinctive claims would be weakened if any of the following replicated results occurred:

1. state-only and item-reactivation models equal or outperform the rendering model on held-out content, viewpoint, and transition data;
2. source attribution cannot vary independently of scene vividness or content under adequately powered lucid-dream protocols;
3. receiver-relative PWD features add no information beyond capacity-matched phase, power, rate, waveform, and prediction-error baselines;
4. perturbations targeted to predicted receiver windows fail to cause selective downstream scene changes;
5. cross-episode cueing does not alter recombination beyond base rates and waking-incubation effects;
6. the proposed artificial architecture provides no benefit over exact or generative replay after parameter and compute matching; or
7. results depend on one sleep stage, laboratory, individual, or scoring scheme and fail preregistered replication;
8. source and vividness cannot be separately anchored with a stable full-rank observation map;
9. PWD coefficients are conditionally collinear with the strongest phase, power, waveform, state, and generic prediction-error alternatives;
10. apparent performance disappears when whole participants rather than event rows are held out;
11. source probabilities are materially miscalibrated despite acceptable discrimination; or
12. conclusions reverse across plausible MNAR report-sensitivity settings.
13. the reference application cannot improve hybrid generalization, constrain impossible transitions, or calibrate source probabilities on a later unopened test after all repairs are frozen; or
14. a receiver-conditioned generated-replay model continues to tie a simpler exact-replay comparator when storage, compute, and environment complexity are matched at the intended application scale.

Negative results would refine which components belong in the operator. They would not justify substituting a compressed interpretation for the source-faithful SAN claim. The Apex alignment procedure requires identifying whether a test addressed the same operation, scale, state, receiver, and outcome before drawing a semantic conclusion.

9. Red-team requirements without strawman compression

A challenge to this paper is admissible only after reconstructing its strongest causal argument:

> *Dreams are state-conditioned recurrent scenes assembled from learned sensory, mnemonic, affective, bodily, and action relations. Their content becomes operational because it changes receiver populations and later transitions. Source attribution and conscious access are separable from scene generation. NAPOT and PWD add a proposed receive-transform-project and receiver-relative-difference account that must outperform established alternatives in held-out and causal tests.*

A critique fails before scientific adjudication if it reduces the claim to any of the following:

- dreams predict external future events;
- dreams are only REM sleep;
- memory consolidation requires recalled dreams;
- brain activity is literal computer graphics;

- every oscillation or phase difference is a PWD;
- NAPOT is a localized internal screen or homunculus; or
- a dream report proves its own supernatural interpretation.

The corrected procedure is source recovery, strongest-form reconstruction, scale and terminology translation, external evidence check, then Apex semantic alignment. Established observations, SAN interpretations, and untested extensions remain visibly distinct throughout that process.

10. Ethics and practical boundaries

Dream cueing and lucid-dream experiments can affect sleep quality, emotional content, privacy, and later memory. Studies should exclude unsafe stimulation, monitor distress, allow immediate withdrawal, minimize sleep disruption, and protect dream reports as sensitive mental data. Clinical populations require separate medical protocols. No current evidence justifies unsupervised attempts to suppress broad sensory systems or induce full-dive virtual worlds.

Decoding results should be communicated carefully. Above-chance category inference is not transparent access to a person's entire dream. A future interface that writes or biases dream content would require consent, provenance logs, uncertainty displays, and strict separation between a cue delivered by the device and content generated by the participant.

11. Discussion

Dreams show that conscious scenes need not be continuously dictated by present external input. That does not make them unconstrained. They inherit learned structure, active concerns, body models, semantic relations, and state-dependent physiology. Their transitions can be surprising relative to waking reality while remaining intelligible to the system currently generating them.

The SAN genealogy adds value because it preserves the development of this question across different technologies and levels. The 2011 source asked how distributed partial signals form one predictive multisensory picture. The 2017 discussion explored feedback, hallucination, and recurrent thalamocortical meaning construction. The 2018 discussion proposed render-process-render recurrence. The 2021 recordings separated lucid interface, associative replay, personal-holodeck construction, and recurrent flow. The 2022 work supplied NAPOT and PWD. The 2024 program explicitly linked dreams to prediction, memory, affect, choice, creativity, and self-location. The present formalism translates those stages into measurable variables without pretending that later equations were already present in early notes.

This framework also explains why no single frequency should be expected to solve dreaming. Oscillations can coordinate windows for communication and memory exchange, but content depends on which populations participate, their learned receiver states, the relative timing and waveform of departures, the body and viewpoint model, and the consequences of those departures for the next state. Synchrony can provide coordination; differentiated, receiver-effective departures help specify what changes.

The most decisive evidence will come from dissociations. Can source attribution improve while a vivid scene remains? Can memory reactivation occur without reportable content? Can the same cue change consolidation without changing dream construction, or vice versa? Can a receiver-relative model predict the next dream event after generic phase and prediction-error baselines are exhausted? Can viewpoint be perturbed independently of object content? These questions turn a philosophical topic into an experimental architecture.

Draft 3 adds a prior question: is the variable being tested uniquely connected to its intended meaning under the measurement design? The synthetic fixtures show why predictive accuracy alone cannot answer that question. A collapsed source/vividness map retained nonzero decoding performance but failed the rank condition. A row-level split looked far better than a participant-held-out split because it learned participant-specific relations. Conversely, the PWD coefficients were recovered accurately only after ordinary signal features were projected out, and the same pipeline returned essentially zero improvement when the true PWD effect was removed. These results validate the gate and its refusal logic, not the biological hypothesis.

Draft 4 adds the missing software question: does the proposed comparison exist as an inspectable application with the same budget, partitions, outputs, and refusal rules for every arm? It now does. The result is informative precisely because it is mixed. Receiver conditioning and PWD carry planted information in the positive world rule, the PWD-null fixture returns no benefit, shuffled replay destroys retention, and row splitting produces a large false advantage. At the same time, exact replay ties the proposed generated-replay arm, hybrid generalization remains modest, impossible transitions remain common, and source calibration misses its diagnostic limit. The next application gate must repair those specific failures without reopening this development partition.

Draft 5 shows why repair must remain a comparison rather than a victory lap. Constrained decoding, calibration, hybrid recombination, and seed stability improve under the frozen implementation, but the compact factorized receiver-aware model solves the transparent generator exactly and at far lower storage cost. More importantly, the nominal PWD-null comparison changes both PWD access and the candidate-set learner, so its large apparent gain is confounded. The correct response is to preserve the result, repair the contrast, and make the next generator harder and less aligned with either model's factorization.

Draft 6 performs that repair and shows why a harder benchmark is scientifically useful. The old factorized receiver architecture falls from perfect accuracy to a median of 0.040, while the relational primary reaches 0.128 and a true one-variable PWD comparison produces a positive median difference of 0.038. Those relative gains coexist with decisive adverse results: three of four absolute gates fail, the impossible-transition rate remains 0.831, the most difficult families remain poorly predicted, the true PWD null crosses its frozen absolute bound on at least one seed, and row splitting remains severely optimistic. The result narrows the engineering problem without elevating a synthetic feature effect into evidence about dreams, neurons, or consciousness.

Draft 7 then separates null calibration from positive evaluation. That design succeeds methodologically: the multiplier is chosen without positive outcomes, the selected full weight passes a second independent null, the full and ablated arms share the estimator and calibration objects, and the future partition remains untouched. The negative result is architectural. The richer evidence table dilutes rather than reconstructs the cross-coupled rule, yields almost no positive PWD difference, and underperforms the simpler retained models at greater storage cost. Null safety is necessary for a feature-specific claim, but it is not sufficient for useful positive identification.

12. Conclusion

Dreaming is not merely offline noise, a single rhythm, a delayed report, or a replayed recording. It is a state-dependent problem of constructing a usable world from internally weighted evidence. Predictive Neural Rendering names the proposed operation only after that problem has been established: learned neural systems reactivate and recombine partial relations; current sleep state changes precision and access; receiver-specific transformations produce consequential differences; recurrent exchange constructs a situated scene; and source attribution determines whether the scene is recognized as internally generated.

The theory is intentionally exposed to failure. Its rendering variables must outperform simpler state and replay models. PWD must add measurable receiver-relative value beyond phase, rate, power, waveform, and generic prediction error. NAPOT must yield causal predictions about cross-population transformation rather than functioning as imagery. Lucidity, self-location, memory incorporation, and later report must dissociate in the ways the architecture predicts. Before any of those results receive a mechanistic label, the intended latent variable must satisfy its anchor and rank conditions, survive participant-held-out testing, remain calibrated, and retain its conclusion across declared report-missingness sensitivities. Those tests define the next step from a dated conceptual genealogy to a cumulative neuroscience program.

The Draft 4 application made one part of that program concrete and recorded why it was not finished. Draft 5 then repaired constrained decoding, calibration, hybrid recombination, and multi-seed stability without erasing those earlier outcomes. Its repaired primary arm reached median exact accuracy 0.915, median hybrid accuracy 0.756, zero impossible transitions, and median source ECE 0.00078 across ten development seeds. Those values did not justify opening the future test: a compact factorized receiver-aware model scored 1.000 throughout, the nominal PWD ablation failed its null with a spurious median gain of 0.299, and random row splitting retained a median optimism gap of 0.750. Those findings motivated, but did not predetermine, the Draft 6 gate.

Draft 6 independently froze a harder four-family development generator and compared the same sixty-key factorized receiver architecture with a relational learner. The factorized model no longer saturated the task, and the relational learner improved on it, but the primary still achieved only 0.128 median exact accuracy, 0.132 median hybrid accuracy, and a 0.831 median impossible-transition rate. The PWD-only comparison was correctly isolated, yet its true-null differences still exceeded the strict absolute bound on at least one seed. The manuscript therefore stops at an executed but uncleared development gate. The next honest step is to repair event-specific admissibility and the difficult cross-coupled, role-reversal, and ambiguous-mixture families, then repeat the true one-variable null without opening the sealed future partition.

Draft 7 resolves one part of that request and fails another. The registered null-calibration rule selects the full PWD multiplier, and the independent null remains within its frozen bound. Yet median exact accuracy falls to 0.036, the positive PWD difference falls to 0.0028, all family floors fail, invalid transitions remain common, and the estimator uses more resources than the stronger Draft 6 model. This is an executed adverse development gate, not a reason to open the future partition. The next architecture must recover known planted interactions while preserving the null-calibration discipline and every stronger comparator.

13. Draft 8 bounded formal gate: what passed nulls cannot establish

Draft 7 produced an important asymmetry. Its multiplier-selection procedure passed the registered calibration nulls, and the selected multiplier passed a second set of independent synthetic nulls. Yet the same frozen estimator failed to recover the planted positive effect, performed worse than both retained comparators, failed every family floor, and consumed more storage. Draft 8 does not run a new model to conceal that result. It formalizes the finite design and stopping rule so that a favorable null result cannot be silently promoted into positive identification.

This is a mathematical bookkeeping gate over the executed synthetic record. It is not a mathematical model of dreaming, a proof of biological PWD, or an empirical advance. Its purpose is narrower and useful: to make partition separation, intervention identity, endpoint ordering, adverse-history retention, evidence class, and the closed future seal machine-checkable inside the declared finite model.

13.1 Registered partitions and null-only selection

Let (C), (P), and (N) be the explicitly registered calibration-null, positive-development, and independent-null seed lists. Lean checks their recorded cardinalities, uniqueness, and pairwise separation:

$$|C| = 6, \qquad |P| = 8, \qquad |N| = 8, \qquad C \cap P = C \cap N = P \cap N = \varnothing. \tag{54}$$

These are finite properties of the registered identifiers. They show that the positive fixtures did not share a registered seed with the two null fixtures; they do not establish external custody, laboratory independence, or biological generalization.

The multiplier grid is represented in basis points as

$$\Lambda_{bp} = \{0, 50, 100, 200, 350, 500, 750, 1000\}. \tag{55}$$

For the frozen Draft 7 eligibility vector, every candidate is eligible, so the registered largest-eligible operator returns 1000 basis points. Lean checks that the returned value belongs to the grid and is no smaller than any registered candidate. This proves the finite selection consequence conditional on the encoded eligibility values. The deterministic validator, not Lean, checks that those values and the grid match the frozen specification and results.

13.2 Matched software intervention

The formal arm record contains estimator, row, shortlist, calibrator, non-PWD-term, PWD-term, and multiplier fields. The full and ablated arms contain 211 non-PWD terms and 43 PWD terms. Replacing the full arm's PWD multiplier with zero yields the ablated arm:

$$E_0(A_{full}) = A_{-PWD}, \qquad 211 + 43 = 254. \tag{56}$$

This equality is the formal counterpart of the registered software comparison. It makes the one declared arm difference explicit. It does not prove that the software feature is identical to a neural phase-wave differential, nor does it turn a synthetic ablation into causal neuroscience.

13.3 Null safety and positive identification are different gates

The recorded calibration maxima are 6,944 and 8,333 millionths, both below the 15,000-millionth calibration bound. The recorded independent-null maxima are 6,944 and 10,417 millionths, both below the 20,000-millionth bound. Lean checks these finite inequalities.

The positive result points in the opposite evidentiary direction. The median positive PWD gain is 2,778 millionths, below its frozen 20,000-millionth floor. Both retained comparators exceed the Draft 7 primary on exact and hybrid medians, and all four family medians remain below their frozen floors. The resulting vector is

$$\begin{aligned} N_{cal} &= 1, & N_{ind} &= 1, \\ P_{gain} &= 0, & P_{pred} &= 0, \\ P_{comp} &= 0, & P_{family} &= 0. \end{aligned} \tag{57}$$

Define null specificity as $N_7 = N_{\text{cal}} \wedge N_{\text{ind}}$, and define positive identification as $P_7 = P_{\text{gain}} \wedge P_{\text{pred}} \wedge P_{\text{comp}} \wedge P_{\text{family}}$. For the frozen gate vector, Lean proves

$$N_7 = 1 \quad \wedge \quad P_7 = 0. \tag{58}$$

Equation 58 is a negative-identification theorem about this recorded gate, not a general theorem that null controls can never inform mechanism claims. Here, the passed nulls constrain false positive behavior, while the failed positive endpoints block a claim that the estimator successfully identifies the planted relation. The correct interpretation is therefore stable: the null-calibration discipline worked; the estimator did not.

13.4 Comparator, resource, and adverse-history invariants

Lean checks the recorded exact and hybrid ordering for the Draft 7 primary, retained Draft 6 relational comparator, and retained factorized comparator. It also checks that the 5,202-key Draft 7 estimator is larger than the sixty-key factorized estimator and that its lower storage estimate exceeds the upper recorded estimates for both retained models. These are endpoint and implementation inequalities. They do not establish a universal complexity theorem or a biological mapping from bytes to cognition.

Adverse disclosure is additive:

$$\text{49 inherited rows} + \text{72 Draft 7 rows} = \text{121 retained rows.} \tag{59}$$

Lean proves the arithmetic. The deterministic validator separately proves that the first 49 CSV rows remain byte-for-byte equivalent as parsed records and that the source file hashes remain frozen. This split matters: a theorem about counts cannot authenticate row content, while a content validator does not replace the theorem about the declared ledger relation.

13.5 Seal and evidence-class boundary

The formal state machine maps a closed seal to `execution permitted = false`. It also maps the Draft 7 record to synthetic-development evidence while returning false for human dream evidence, biological PWD validation, causal neural perturbation, artificial-consciousness inference, and publication authorization:

$$\begin{aligned} \text{Support}_{D7}(e) &= \begin{cases} 1, & e = \text{synthetic development}, \\ 0, & e \in \mathcal{U}_{\text{unsupported}}, \end{cases} \\ \mathcal{U}_{\text{unsupported}} &= \{\text{human dream, biological PWD, neural cause,} \\ &\qquad \text{artificial consciousness, publication}\}. \end{aligned} \tag{60}$$

This is a type boundary encoded from the declared evidence contract. It prevents a machine-checked finite result from being connected to a claim class for which no premise was supplied. It does not decide the truth of the broader SAN hypotheses.

13.6 Proof status and compatibility boundary

The formal leaf contains thirty-five theorems. Its claim signature separates a green finite kernel from a yellow source-transcription bridge. Compatibility edges allow exact assembly only inside the finite synthetic gate, require review where JSON/CSV values are mapped into Lean constants, and explicitly block assembly into human-dream, biological-PWD, consciousness, future-test, or publication claims.

Formal block	What is checked	What remains outside the proof
seed registration	counts, uniqueness, pairwise disjointness	custody and external independence
multiplier rule	grid membership and maximal eligible value	authentication of upstream eligibility data
intervention identity	equality after multiplier erasure	biological identity of the PWD feature
endpoint vector	recorded finite inequalities	population inference and external validity
disclosure	count additivity	row-content authentication, checked separately
evidence boundary	closed seal and unsupported claim classes	the truth of broader neuroscience hypotheses

Draft 8 therefore closes a formal interpretation gate without advancing the empirical gate. The future partition remains closed. The next empirical architecture still must recover the planted positive relation, exceed the retained comparators, satisfy all family and validity floors, preserve null specificity, and be frozen before any separately authorized future test.

14. Draft 9 structural-program repair: recovering the planted relation without erasing the adverse gate

The preceding gate exposed a precise engineering problem. A null-safe estimator is not useful if it cannot reconstruct a relation known to exist in the synthetic world. The next repair therefore had to recover the planted relation, preserve the one-variable PWD comparison, outperform the two strongest retained models, remain quiet under an independent PWD null, satisfy family and validity floors, and leave the future partition closed. It also had to retain every failed result rather than replacing the history with a favorable summary.

Draft 9 tests that question with a deliberately aligned candidate library. For each of the four frozen generator families, training data choose among four declared programs: branch 0 or branch 1, each with the PWD coefficient present or absent. This library was written after inspecting the frozen synthetic generator. It is therefore a known-ground-truth system-identification test, not architecture-blind generalization. The purpose is to ask whether the previously missed relation becomes recoverable under an explicit structural representation while null specificity and matched intervention identity remain intact.

14.1 Training-selected structural programs

For family (f), let the declared library be

$$\mathcal{C}_f = \{(b,p) : b \in \{0,1\},\ p \in \{0,1\}\}. \tag{61}$$

Training-only exact relation accuracy selects

$$c_f^* = \arg\max_{c \in \mathcal{C}_f} A_{train}(c), \tag{62}$$

with a frozen tie-break favoring the lower PWD coefficient and then the lower branch identifier. The tie-break prevents an equal-performing PWD program from being preferred merely because it contains the feature under test. All 480 seed-family-candidate rows are retained: 120 selected rows and 360 rejected rows. Across all eight positive-development seeds, every family selects one modal program with frequency 1.0. F1, F2, and F3 select branch 0 with the PWD coefficient; F4 selects branch 1 with the PWD coefficient.

The full and ablated arms share the same selected-program object, training rows, candidate library, and source calibrator. Their only runtime difference is

$$\hat{y}_{full} = g(x, c_f^*, \lambda), \qquad \hat{y}_{-PWD} = g(x, c_f^*, 0). \tag{63}$$

Six new null-calibration seeds select the largest eligible value from the frozen grid $\{0,1\}$. Both exact and hybrid null gains are zero at $\lambda = 1$, so the registered rule selects 1. Eight separate positive seeds then evaluate the program, and eight further null seeds test the frozen multiplier. No result from the positive or independent-null partitions can alter the library, tie-break, multiplier rule, thresholds, or arms.

14.2 Positive recovery, comparators, and independent null

The aligned program recovers most of the planted transition relation. Median exact accuracy is 0.8601, with a minimum of 0.8500 across seeds. The PWD-ablated version reaches 0.2226, producing a median matched PWD gain of 0.6340. The retained Draft 6 relational comparator reaches 0.1253 and the retained factorized comparator reaches 0.0438. The primary program therefore exceeds them by 0.7347 and 0.8163, respectively. Its median impossible-transition rate is 0, median ten-bin source ECE is 0.0907, and its serialized selected-program representation is 1,630 bytes.

Family medians are 0.9292 for F1, 0.9014 for F2, 0.8833 for F3, and 0.7347 for F4, all above their frozen family floors. On the eight independent PWD-null seeds, exact and hybrid full-minus-ablated gains are identically zero. The principal finite result vector is therefore

$$\begin{aligned} P_{exact} &= 1, & P_{valid} &= 1, & P_{cal} &= 1, \\ P_{comp} &= 1, & P_{PWD} &= 1, & P_{family} &= 1, \\ N_{ind} &= 1, & P_{hybrid} &= 0. \end{aligned} \tag{64}$$

This is a substantial recovery inside the declared synthetic system. It does not show that biological dream transitions follow the same formulas, that a neural PWD has been measured, or that the candidate library generalizes to an unknown architecture. Training labels select among formulas constructed from the known generator grammar; prediction itself reads no target or allowed-target field.

14.3 The failed hybrid floor remains governing

The frozen median hybrid floor is 0.9000. The observed median is 0.8625, so Draft 9 remains blocked. The failure cannot be removed by pointing to the stronger exact result, the large PWD gain, or the passed independent null.

After execution, the reason for the mismatch is interpretable. The four frozen families choose their modal hidden branch with probabilities 0.92, 0.90, 0.88, and 0.74. A deterministic program that selects one branch per family therefore has an equal-family expected modal-branch ceiling of

$$\bar{q}_{mode} = \frac{0.92 + 0.90 + 0.88 + 0.74}{4} = 0.86. \tag{65}$$

The observed hybrid median of 0.8625 lies near that value. This does not retrospectively change the 0.9000 threshold or convert the failed gate into a pass. It reveals a design mismatch between a deterministic modal-branch model class and a threshold above that class's expected branch ceiling. A later gate must resolve the mismatch before outcomes, either by introducing a declared probabilistic mixture predictor that can represent branch uncertainty or by defining performance relative to a precomputed observable ceiling. Draft 9 itself remains frozen and adverse on the registered hybrid endpoint.

The row-split diagnostic also remains adverse. Its median optimism gap is 0.6875. This reiterates the existing rule that event rows nested within agents cannot be treated as independent participants.

14.4 Losslessness, strongest-form interpretation, and stopping rule

The Draft 9 adverse ledger begins with all 121 Draft 7 rows in their original parsed order. It then retains 360 rejected program candidates, twelve null-calibration rows, the failed hybrid decision, the adverse row-split decision, eight independent-null rows, and eight row-split rows, for 511 total rows. Primary and replay executions produce 22 files each, with 21 payload files byte-identical before their manifests are compared; the manifests are also identical. The existing future partition remains closed and unopened.

The strongest source-faithful SAN proposition is not that one handcrafted program proves dreaming. It is that dreams can be modeled as state-conditioned recurrent renderings assembled from learned relations; that source attribution and conscious access are separable from scene construction; and that receiver-relative differences should have measurable consequences for later transitions beyond generic phase, power, rate, waveform, and prediction-error variables. Draft 9 tests only a synthetic prerequisite: whether a declared representation can recover a planted receiver/PWD relation under matched ablation and null controls.

Three compressions are therefore rejected. First, the Draft 7 estimator failure did not show that PWD or Predictive Neural Rendering was false; it showed that one frozen estimator did not identify its planted target. Second, the Draft 9 recovery does not establish the biological mechanism; it shows that explicit structural representation solves most of the declared synthetic task. Third, the missed hybrid floor is neither erased nor inflated into a verdict on SAN. It is a registered engineering failure whose source can guide a separately frozen repair.

Draft 9 closes as `EXECUTED_MIXED_DEVELOPMENT_ADVANCEMENT_BLOCKED`. The next permissible gate is a new, pre-outcome probabilistic-mixture or ceiling-aware development design on fresh synthetic seeds. It must preserve the current program, both retained comparators, matched PWD ablation, independent-null specificity, all 511 adverse rows, whole-agent inference, and the closed future seal. The future test must not be opened.

15. Draft 10 probabilistic latent-mixture and ceiling-aware gate

Draft 9 established that a structure-aligned program could recover the planted synthetic relation, yet its deterministic one-branch-per-family architecture missed the frozen 0.9000 hybrid floor. That result remains failed. Draft 10 does not lower the floor or rename the failure. It asks a new question on fresh seeds: can a preregistered probabilistic model represent the latent branch uncertainty, achieve the declared modal ceiling, assign calibrated probability to both allowed branches, preserve a matched PWD intervention and independent null, and retain every prior adverse result?

The answer is yes inside this deliberately generator-aligned synthetic system. This is a bounded application result, not biological evidence about dreaming.

15.1 Why a probability distribution is required

The frozen generator draws one of two latent branches for every event. Its branch-one probabilities are

$$q = (0.08, 0.10, 0.12, 0.74), \tag{66}$$

so the family-specific modal probabilities are

$$m = (0.92, 0.90, 0.88, 0.74). \tag{67}$$

With equal family frequency, the preregistered maximum expected accuracy of a deterministic family-modal predictor is

$$\bar{m} = \frac{1}{4}\sum_{f=1}^{4} m_f = 0.86. \tag{68}$$

The corresponding oracle branch entropy is

$$H(B \mid F) = \frac{1}{4}\sum_{f=1}^{4}[-q_f \log q_f - (1-q_f)\log(1-q_f)] = 0.38596 \text{ nats}. \tag{69}$$

Equations 68 and 69 were frozen before Draft 10 outcomes. They expose two distinct questions. Maximum-a-posteriori accuracy asks whether the most probable branch was selected. Proper scoring asks whether the model assigned the right probability to both possible branches. A probability model can be well specified even though unavoidable branch draws prevent deterministic accuracy from approaching one.

15.2 Frozen mixture and matched PWD intervention

For each family, training chooses $p \in \{0, 1\}$, maps unambiguous rows to branch zero or one, and estimates the branch-one probability with Laplace constant $\alpha = 1$:

$$\hat{q}_f = \frac{n_{f,1} + \alpha}{n_{f,0} + n_{f,1} + 2\alpha}. \tag{70}$$

The selected coefficient minimizes training mixture negative log likelihood, with ties favoring the lower PWD coefficient. For event x, branch relations $r_{f,0}(x, p)$ and $r_{f,1}(x, p)$, and runtime multiplier λ, the prediction is

$$P(y \mid x, f) = (1 - \hat{q}_f)\mathbf{1}[y = r_{f,0}(x, p; \lambda)] + \hat{q}_f \mathbf{1}[y = r_{f,1}(x, p; \lambda)]. \tag{71}$$

The full and ablated arms share one mixture object, the same learned weights, rows, candidate formulas, and source calibrator. Their only runtime difference is $\lambda = 1$ versus $\lambda = 0$. Six fresh PWD-null seeds select λ without positive outcomes. Eight fresh positive seeds and eight further null seeds then evaluate the frozen result. The runner reads no target or allowed-target field during prediction.

15.3 Proper scores and ceiling attainment

The complete-token mixture negative log likelihood and multiclass Brier score are

$$\mathrm{NLL} = -\frac{1}{n} \sum_{i=1}^{n} \log P(y_i \mid x_i), \tag{72}$$

$$\mathrm{Brier} = \frac{1}{n} \sum_{i=1}^{n} \sum_{k} [P(k \mid x_i) - \mathbf{1}(y_i = k)]^2. \tag{73}$$

Top-two coverage records whether the observed token lies in the two-branch support. Ceiling attainment is the observed maximum-a-posteriori exact accuracy divided by the preregistered modal ceiling; it is not clipped at one because finite samples can exceed their expectation.

On the eight positive seeds, median exact accuracy is 0.86146 and the minimum is 0.84375. Median ceiling attainment is 1.00170, with range 0.98110-1.01340. Family ceiling-attainment ratios are 1.00393, 0.98765, 1.00063, and 1.01164 for F1-F4. Median NLL is 0.38804 nats, only 0.00208 above the registered oracle entropy. Median multiclass Brier score is 0.23107, top-two coverage is 1.000, and maximum training-weight error across positive seed-family fits is 0.03426.

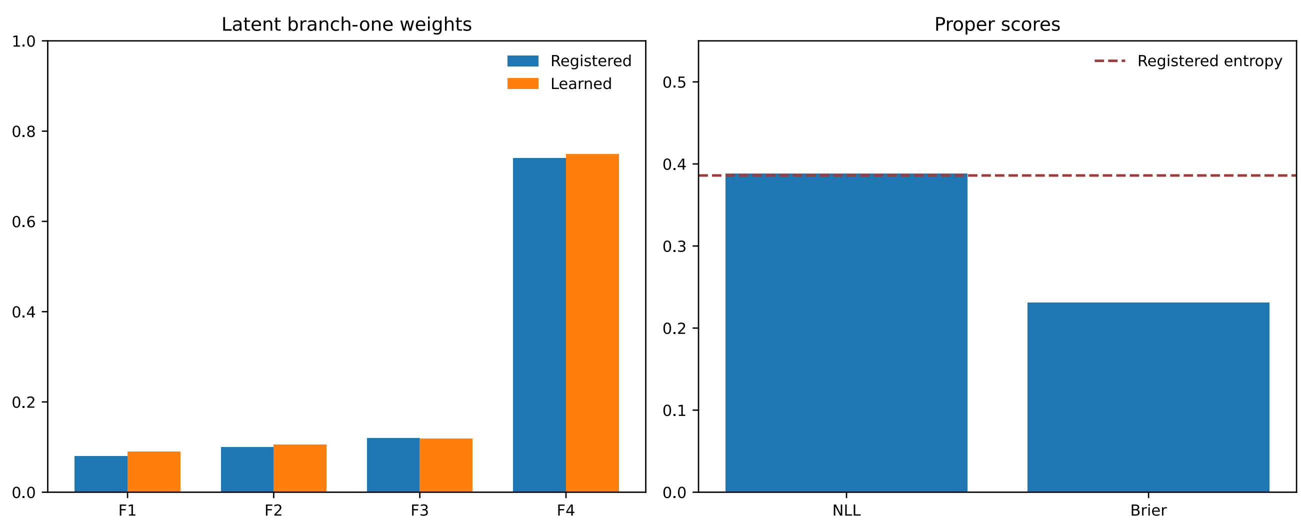

Figure 8. Probabilistic mixture calibration. The learned training-only branch weights track the frozen generator probabilities, while the complete-token NLL remains close to the registered branch entropy.

15.4 Comparators, PWD specificity, and retained adverse evidence

The Draft 10 and retained Draft 9 maximum-a-posteriori programs both reach median exact accuracy 0.86146 on the fresh seeds. The retained Draft 6 relational model reaches 0.11944 and the factorized model 0.03854. Draft 10 therefore preserves deterministic recovery while adding calibrated uncertainty rather than claiming an artificial accuracy increase beyond the stochastic ceiling.

The full-minus-ablated median exact gain is 0.64201. The median improvement in NLL is 14.87663 nats because ablation often removes the observed target from the two-branch support. On all eight independent PWD-null seeds, exact, hybrid, and NLL differences are zero. Median impossible-transition rate is zero and median source ECE is 0.12903.

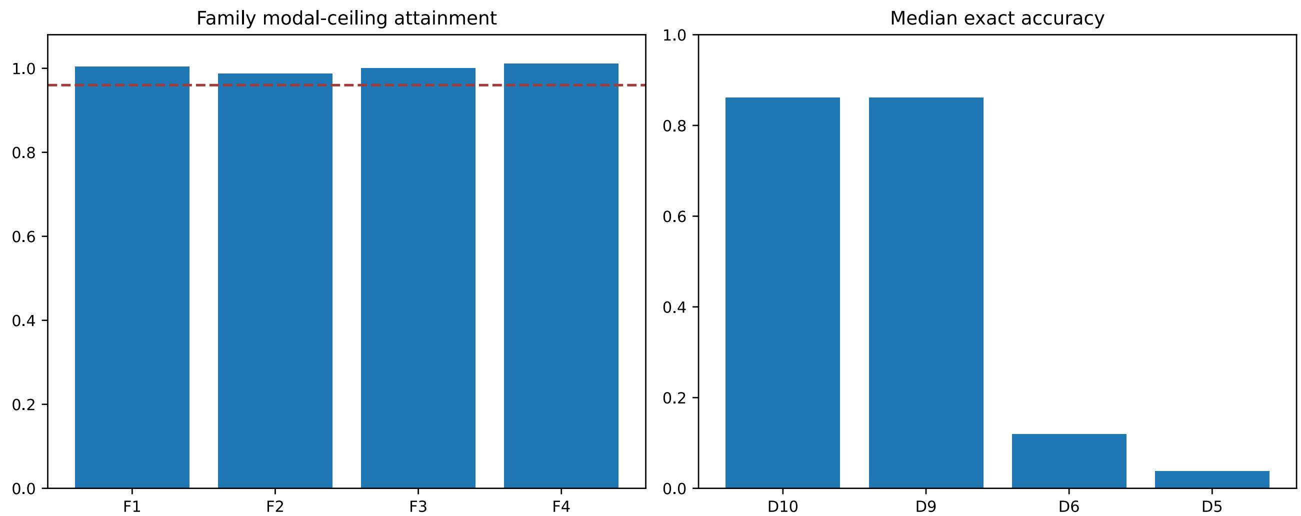

Figure 9. Ceiling-aware and comparator results. Draft 10 reaches the preregistered family modal ceilings while retaining Draft 9 deterministic performance and exceeding the older comparators.

All 21 new advancement endpoints pass. That does not erase adverse history. The Draft 10 ledger begins with all 511 Draft 9 rows in their prior order and retains the failed Draft 9 0.9000 hybrid floor, 120 rejected coefficient candidates, twelve null-calibration outcomes, eight independent-null results, and eight row-split diagnostics. The resulting ledger contains 661 rows. Row-split optimism remains adverse. Primary and replay executions each contain 23 files and are byte-identical.

15.5 What the passed gate permits

Draft 10 supports one constructive claim: the reference application now contains a preregistered probabilistic model that recovers a known planted receiver/PWD relation up to its declared stochastic ceiling, assigns calibrated mass to latent alternatives, outperforms retained simpler models on the same fresh fixtures, passes matched PWD null controls, and reproduces exactly.

It does not support the claim that human dreams have been decoded, that biological PWD has been observed, that neurons implement these formulas, that NAPOT has been causally validated, or that the synthetic application is conscious. The candidate formulas and registered branch probabilities intentionally follow the frozen generator. Architecture-blind evaluation, independent implementation, real sleep data, measured neural variables, prospective perturbation, and participant-level replication remain future empirical work.

The historical Draft 9 floor remains `ADVERSE_RETAINED_NOT_REOPENED`. Draft 10 passes a different model-class gate whose proper-score and ceiling-aware endpoints were frozen before outcomes. This distinction is essential: a new question can be answered without rewriting an old answer.

15.6 Bounded final-preprint decision

The manuscript now qualifies as a bounded theory-and-methods preprint with an executable synthetic reference application. The word *bounded* carries the scientific burden: the release may report the source genealogy, theory, mathematical definitions, application architecture, negative gates, successful probabilistic repair, exact replay, and falsifiers. It must label the application as generator-aligned synthetic development and must not present it as biological confirmation or an unopened future-test result.

No successor draft is required to make that bounded preprint internally coherent. A successor is required for any stronger evidence claim, including architecture-blind generalization, external replication, human dream prediction, biological PWD, causal neural rendering, or future-partition performance. The future partition remains closed and unopened.

16. Final bounded conclusion

Dreams create a tractable rendering problem because a situated, affective, action-relevant world can be experienced while ordinary external constraint is altered. The SAN proposal explains that problem with learned relations, receiver-dependent transformation, recurrent projection, source attribution, and consequential differences rather than a localized viewer or one universal rhythm. The accompanying application now demonstrates that this operation can be specified, falsified, repaired, probability-calibrated, and replayed in a declared synthetic world. The adverse gates remain part of the result.

The next scientific burden is external: architecture-blind benchmarks, independent implementation, participant-held-out sleep studies, measured receiver-relative neural variables, and prospective perturbations that distinguish PWD and NAPOT from established alternatives. Until those bridges exist, the application is a rigorous synthetic reference implementation and the neuroscience remains a falsifiable research program. The future final-test partition remains closed.

First-party SAN sources

The historical sources are catalogued with dates, hashes, authorship, privacy, and boundary fields in the Draft 2 genealogy ledger and source freeze. Draft 7 preserves that genealogy unchanged and adds only 2026 mathematical conditions, typed application code, deterministic synthetic fixtures, confound controls, development repairs, a harder generator family, registered null calibration, and method sources. These sources establish genealogy and authorial meaning; they are not substituted for independent experimental evidence.

References

1. Aserinsky, E., & Kleitman, N. (1953). Regularly occurring periods of eye motility, and concomitant phenomena, during sleep. *Science, 118*, 273-274. https://doi.org/10.1126/science.118.3062.273
2. Dement, W., & Kleitman, N. (1957). The relation of eye movements during sleep to dream activity. *Journal of Experimental Psychology, 53*, 339-346. https://doi.org/10.1037/h0048189
3. Maquet, P., Péters, J., Aerts, J., Delfiore, G., Degueldre, C., Luxen, A., & Franck, G. (1996). Functional neuroanatomy of human rapid-eye-movement sleep and dreaming. *Nature, 383*, 163-166. https://doi.org/10.1038/383163a0
4. Siclari, F., Baird, B., Perogamvros, L., et al. (2017). The neural correlates of dreaming. *Nature Neuroscience, 20*, 872-878. https://doi.org/10.1038/nn.4545
5. Horikawa, T., Tamaki, M., Miyawaki, Y., & Kamitani, Y. (2013). Neural decoding of visual imagery during sleep. *Science, 340*, 639-642. https://doi.org/10.1126/science.1234330
6. Horikawa, T., & Kamitani, Y. (2017). Generic decoding of seen and imagined objects using hierarchical visual features. *Nature Communications, 8*, 15037. https://doi.org/10.1038/ncomms15037
7. Dresler, M., Koch, S. P., Wehrle, R., et al. (2011). Dreamed movement elicits activation in the sensorimotor cortex. *Current Biology, 21*, 1833-1837. https://doi.org/10.1016/j.cub.2011.09.029
8. Dresler, M., Wehrle, R., Spoormaker, V. I., et al. (2012). Neural correlates of dream lucidity obtained from contrasting lucid versus non-lucid REM sleep: A combined EEG/fMRI case study. *Sleep, 35*, 1017-1020. https://doi.org/10.5665/sleep.1974

9. Baird, B., Mota-Rolim, S. A., Dresler, M., et al. (2018). Smooth tracking of visual targets distinguishes lucid REM sleep dreaming and waking perception from imagination. *Nature Communications, 9*, 3298. https://doi.org/10.1038/s41467-018-05547-0
10. Konkoly, K. R., Appel, K., Chabani, E., et al. (2021). Real-time dialogue between experimenters and dreamers during REM sleep. *Current Biology, 31*, 1417-1427.e6. https://doi.org/10.1016/j.cub.2021.01.026
11. Konkoly, K. R., Al-Youssef, S., Mazurek, C. Y., Mallett, R., Morris, D. J., Gales, A., Arnulf, I., Oudiette, D., & Paller, K. A. (2026). Using real-time reporting to investigate visual experiences in dreams. *Journal of Cognitive Neuroscience, 38*(2), 365-380. https://doi.org/10.1162/JOCN.a.107
12. Rasch, B., Büchel, C., Gais, S., & Born, J. (2007). Odor cues during slow-wave sleep prompt declarative memory consolidation. *Science, 315*, 1426-1429. https://doi.org/10.1126/science.1138581
13. Rudoy, J. D., Voss, J. L., Westerberg, C. E., & Paller, K. A. (2009). Strengthening individual memories by reactivating them during sleep. *Science, 326*, 1079. https://doi.org/10.1126/science.1179013
14. Wamsley, E. J., Tucker, M., Payne, J. D., Benavides, J. A., & Stickgold, R. (2010). Dreaming of a learning task is associated with enhanced sleep-dependent memory consolidation. *Current Biology, 20*, 850-855. https://doi.org/10.1016/j.cub.2010.03.027
15. Wamsley, E. J., & Stickgold, R. (2019). Dreaming of a learning task is associated with enhanced memory consolidation: Replication in an overnight sleep study. *Journal of Sleep Research, 28*, e12749. https://doi.org/10.1111/jsr.12749
16. Schönauer, M., Alizadeh, S., Jamalabadi, H., Abraham, A., Pawlizki, A., & Gais, S. (2017). Decoding material-specific memory reprocessing during sleep in humans. *Nature Communications, 8*, 15404. https://doi.org/10.1038/ncomms15404
17. Staresina, B. P., Bergmann, T. O., Bonnefond, M., et al. (2015). Hierarchical nesting of slow oscillations, spindles and ripples in the human hippocampus during sleep. *Nature Neuroscience, 18*, 1679-1686. https://doi.org/10.1038/nn.4119
18. Helfrich, R. F., Mander, B. A., Jagust, W. J., Knight, R. T., & Walker, M. P. (2019). Bidirectional prefrontal-hippocampal dynamics organize information transfer during sleep in humans. *Nature Communications, 10*, 3572. https://doi.org/10.1038/s41467-019-11444-x
19. Sterpenich, V., Perogamvros, L., Tononi, G., & Schwartz, S. (2020). Fear in dreams and in wakefulness: Evidence for day/night affective homeostasis. *Human Brain Mapping, 41*, 840-850. https://doi.org/10.1002/hbm.24843
20. Cai, D. J., Mednick, S. A., Harrison, E. M., Kanady, J. C., & Mednick, S. C. (2009). REM, not incubation, improves creativity by priming associative networks. *Proceedings of the National Academy of Sciences, 106*, 10130-10134. https://doi.org/10.1073/pnas.0900271106
21. Horowitz, A. H., et al. (2023). Targeted dream incubation at sleep onset increases post-sleep creative performance. *Scientific Reports, 13*, 7319. https://doi.org/10.1038/s41598-023-31361-w
22. Blanke, O., Ortigue, S., Landis, T., & Seeck, M. (2002). Stimulating illusory own-body perceptions. *Nature, 419*, 269-270. https://doi.org/10.1038/419269a
23. Ehrsson, H. H. (2007). The experimental induction of out-of-body experiences. *Science, 317*, 1048. https://doi.org/10.1126/science.1142175
24. Lenggenhager, B., Tadi, T., Metzinger, T., & Blanke, O. (2007). Video ergo sum: Manipulating bodily self-consciousness. *Science, 317*, 1096-1099. https://doi.org/10.1126/science.1143439
25. Hobson, J. A., & McCarley, R. W. (1977). The brain as a dream state generator: An activation-synthesis hypothesis of the dream process. *American Journal of Psychiatry, 134*, 1335-1348. https://doi.org/10.1176/ajp.134.12.1335
26. Hobson, J. A., & Friston, K. J. (2012). Waking and dreaming consciousness: Neurobiological and functional considerations. *Progress in Neurobiology, 98*, 82-98. https://doi.org/10.1016/j.pneurobio.2012.05.003
27. Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience, 2*, 79-87. https://doi.org/10.1038/4580
28. Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience, 11*, 127-138. https://doi.org/10.1038/nrn2787
29. Revonsuo, A. (2000). The reinterpretation of dreams: An evolutionary hypothesis of the function of dreaming. *Behavioral and Brain Sciences, 23*, 877-901. https://doi.org/10.1017/S0140525X00004015
30. Windt, J. M. (2010). The immersive spatiotemporal hallucination model of dreaming. *Phenomenology and the Cognitive Sciences, 9*, 295-316. https://doi.org/10.1007/s11097-010-9163-1
31. Llewellyn, S. (2013). Such stuff as dreams are made on? Elaborative encoding, the ancient art of memory, and the hippocampus. *Behavioral and Brain Sciences, 36*, 589-607. https://doi.org/10.1017/S0140525X12003135
32. Demirel, Ç., Gott, J., Appel, K., et al. (2025). Electrophysiological correlates of lucid dreaming: Sensor and source level signatures. *The Journal of Neuroscience, 45*, e2237242025. https://doi.org/10.1523/JNEUROSCI.2237-24.2025
33. Temel Başer, Z., Uzer, G., Şentürk, H., et al. (2026). Bridging dream and wake states: A new tool for consciousness research. *Neuroscience of Consciousness, 2026*(1), niag038. https://doi.org/10.1093/nc/niag038
34. Khemakhem, I., Kingma, D. P., Monti, R. P., & Hyvärinen, A. (2020). Variational autoencoders and nonlinear ICA: A unifying framework. *Proceedings of Machine Learning Research, 108*, 2207-2217. https://proceedings.mlr.press/v108/khemakhem20a.html
35. Saravanan, V., Berman, G. J., & Sober, S. J. (2020). Application of the hierarchical bootstrap to multi-level data in neuroscience. *Neuron, Behavior, Data Analysis, and Theory, 3*(5). https://pmc.ncbi.nlm.nih.gov/articles/PMC7906290/
36. Varoquaux, G. (2018). Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage, 180*, 68-77. https://doi.org/10.1016/j.neuroimage.2017.06.061